

LAB 5

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REG NO: 2023-BSE-022

CLASS: BSE-5A

SUBJECT: CC

LAB TITLE: Java, apt vs apt-get, snap, GUI, Vim on Ubuntu Server

TASK:

Task 1: Discover Missing Command & Install Java Using apt Suggestion Steps (inside the VM terminal):

1. Run the java command to see what the system suggests:

```
Java
hajra@ubuntu-lab:~$ java
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1, or
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install default-jre # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
hajra@ubuntu-lab:~$
```

2. Use the suggested apt command (copy the exact package name suggested by the system):

```
sudo apt install <suggested-package> -y
hajra@ubuntu-lab:~$ sudo apt install openjdk-17-jre-headless
[sudo] password for hajra:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2-data libasound2t64 libpcsclite1
Suggested packages:
  default-jre alsa-utils libasound2-plugins pcsd libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho fonts-woy-microhei
  fonts-woy-zenhei fonts-indic
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2-data libasound2t64 libpcsclite1 openjdk-17-jre-headless
0 upgraded, 8 newly installed, 0 to remove and 20 not upgraded.
Need to get 48.5 MB of archives.
After this operation, 196 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 alsa-topology-conf all 1.2.5.1-2 [15.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libasound2-data all 1.2.11-1ubuntu0.1 [21.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libasound2t64 amd64 1.2.11-1ubuntu0.1 [399 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 alsa-ucm-conf all 1.2.10-1ubuntu5.7 [66.4 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 ca-certificates-java all 20240118 [11.6 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 java-common all 0.75+exp1 [6,796 B]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libpcsclite1 amd64 2.0.3-1build1 [21.4 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openjdk-17-jre-headless amd64 17.0.16+8~us1-0ubuntu1~24.04.1 [48.0 MB]
Fetched 48.5 MB in 7s (6,750 kB/s)
Selecting previously unselected package alsa-topology-conf.
(Reading database ... 85%)
```

3. Verify Java is installed and check version:

```
java -version
hajra@ubuntu-lab:~$ java --version
openjdk 17.0.16 2025-07-15
OpenJDK Runtime Environment (build 17.0.16+8-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.16+8-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
hajra@ubuntu-lab:~$
```

4. Remove the Java package using apt remove (use the same package name you installed):

```
sudo apt remove <suggested-package> -y
hajra@ubuntu-lab:~$ sudo apt remove openjdk-17-jre-headless -y
[sudo] password for hajra:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2-data libasound2t64 libpcsclite1
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  openjdk-17-jre-headless
0 upgraded, 0 newly installed, 1 to remove and 20 not upgraded.
After this operation, 193 MB disk space will be freed.
(Reading database ... 143550 files and directories currently installed.)
Removing openjdk-17-jre-headless:amd64 (17.0.16+8~us1-0ubuntu1~24.04.1) ...
hajra@ubuntu-lab:~$
```

5. Confirm java is no longer available:

```
Java
hajra@ubuntu-lab:~$ java
-bash: /usr/bin/java: No such file or directory
hajra@ubuntu-lab:~$
```

6. Clear the shell's command hash cache (so the shell forgets cached command locations — run as your regular user, no sudo needed):

```
hash -r
```

```

java
hajra@ubuntu-lab:~$ hash -r
hajra@ubuntu-lab:~$ java
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1, or
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install default-jre # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
hajra@ubuntu-lab:~$

```

Task 2: Install & Remove Java Using apt-get (Explicitly)

Steps (inside the VM terminal):

1. Install Java using apt-get

(Choose a common package, e.g., default-jre, or use the same package as in Task 1.)

```
sudo apt-get update
```

```

hajra@ubuntu-lab:~$ sudo apt-get update
[sudo] password for hajra:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
hajra@ubuntu-lab:~$

```

```
sudo apt-get install default-jre -y
```

```

hajra@ubuntu-lab:~$ sudo apt-get install default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  default-jre-headless fonts-dejavu-extra libatk-wrapper-java libatk-wrapper-java-jni libgif7 libxcb-shape0 libxft2 libxv1 libxxf86dga1 openjdk-21-jre
  openjdk-21-jre-headless x11-utils
Suggested packages:
  libnss-mozilla fonts-ipafont-gothic fonts-ipafont-mincho fonts-uwq-microhei | fonts-uwq-zenhei fonts-indic mesa-utils
Recommended packages:
  luit
The following NEW packages will be installed:
  default-jre default-jre-headless fonts-dejavu-extra libatk-wrapper-java libatk-wrapper-java-jni libgif7 libxcb-shape0 libxft2 libxv1 libxxf86dga1
  openjdk-21-jre openjdk-21-jre-headless x11-utils
0 upgraded, 13 newly installed, 0 to remove and 20 not upgraded.
Need to get 49.0 MB of archives.
After this operation, 212 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openjdk-21-jre-headless amd64 21.0.8+9~us1-0ubuntu1~24.04.1 [46.4 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 default-jre-headless amd64 2:1.21-75+exp1 [3,094 B]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libgif7 amd64 5.2.2-1ubuntu1 [35.2 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openjdk-21-jre amd64 21.0.8+9~us1-0ubuntu1~24.04.1 [228 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 default-jre amd64 2:1.21-75+exp1 [922 B]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-dejavu-extra all 2.37-8 [1,947 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-shape0 amd64 1.15-1ubuntu2 [6,100 B]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 libxft2 amd64 2.3.6-1build1 [35.9 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 libxv1 amd64 2:1.0.11-1.1build1 [10.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 libxxf86dga1 amd64 2:1.1.5-1build1 [11.6 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 x11-utils amd64 7.7+6build2 [109 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 libatk-wrapper-java all 0.40.0-3build2 [54.3 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/main amd64 libatk-wrapper-java-jni amd64 0.40.0-3build2 [46.4 kB]
Fetched 49.0 MB in 6s (8,240 kB/s)
Selecting previously unselected package openjdk-21-jre-headless:amd64.
(Reading database ... 143263 files and directories currently installed.)
Preparing to unpack .../00-openjdk-21-jre-headless_21.0.8+9~us1-0ubuntu1~24.04.1_amd64.deb ...
Unpacking openjdk-21-jre-headless:amd64 (21.0.8+9~us1-0ubuntu1~24.04.1) ...

```

2. Verify Java version again:

```
java -version
```

```

hajra@ubuntu-lab:~$ java -version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
hajra@ubuntu-lab:~$

```

3. Remove Java using apt-get remove:

```
sudo apt-get remove default-jre -y
```

```

hajra@ubuntu-lab:~$ sudo apt-get remove default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
 alsa-topology-conf alsa-ucm-conf ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libasound2-data libasound2t64 libatk-wrapper-java
  libatk-wrapper-java-jni libgif7 libpcsc-lite1 libxcb-shape0 libxft2 libxv1 libxxf86dga1 openjdk-21-jre openjdk-21-jre-headless x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  default-jre
0 upgraded, 0 newly installed, 1 to remove and 20 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 143707 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
hajra@ubuntu-lab:~$

```

4. Clear the terminal hash cache and confirm Java is missing:

```
hash -r
```

```
java
```

```
hajra@ubuntu-lab:~$ hash -r
hajra@ubuntu-lab:~$ java
-verbose[:class[:module[:gc[:jni]]
    enable verbose output for the given subsystem
--version
    print product version to the error stream and exit
--show-version
    print product version to the output stream and exit
--show-module-resolution
    show module resolution output during startup
-? -h -help
    print this help message to the error stream
--help
    print this help message to the output stream
-X
    print help on extra options to the error stream
--help-extra
    print help on extra options to the output stream
-ea[:<packagename>...[:<classname>]
    enable assertions[:<packagename>...[:<classname>]
-da[:<packagename>...[:<classname>]
    enable assertions with specified granularity
-disableassertions[:<packagename>...[:<classname>]
    disable assertions with specified granularity
-esa | -enableSystemAssertions
    enable system assertions
-dsa | -disableSystemAssertions
    disable system assertions
-agentlib[:<libname>[:<options>]
    load native agent library <libname>, e.g. -agentlib:jdwp
    see also -agentlib:jdwp=help
-agentpath[:<pathname>[:<options>]
    load native agent library by full pathname
-javaagent[:<jarpath>[:<options>]
    load Java programming language agent, see java.lang.instrument
-splash[:<imagepath>]
    show splash screen with specified image
    HI DPI scaled images are automatically supported and used
    if available. The unscaled image filename, e.g. image.ext,
    should always be passed as the argument to the -splash option.
    The most appropriate scaled image provided will be picked up
    automatically.
    See the SplashScreen API documentation for more information
@argument files
    one or more argument files containing options
--disable-@files
    prevent further argument file expansion
--enable-preview
    allow classes to depend on preview features of this release
To specify an argument for a long option, you can use --(name)=<value> or
--(name) <value>.
```

Task 3: apt update vs apt upgrade – Run & Explain Steps (inside the VM terminal):

1. Update the package index

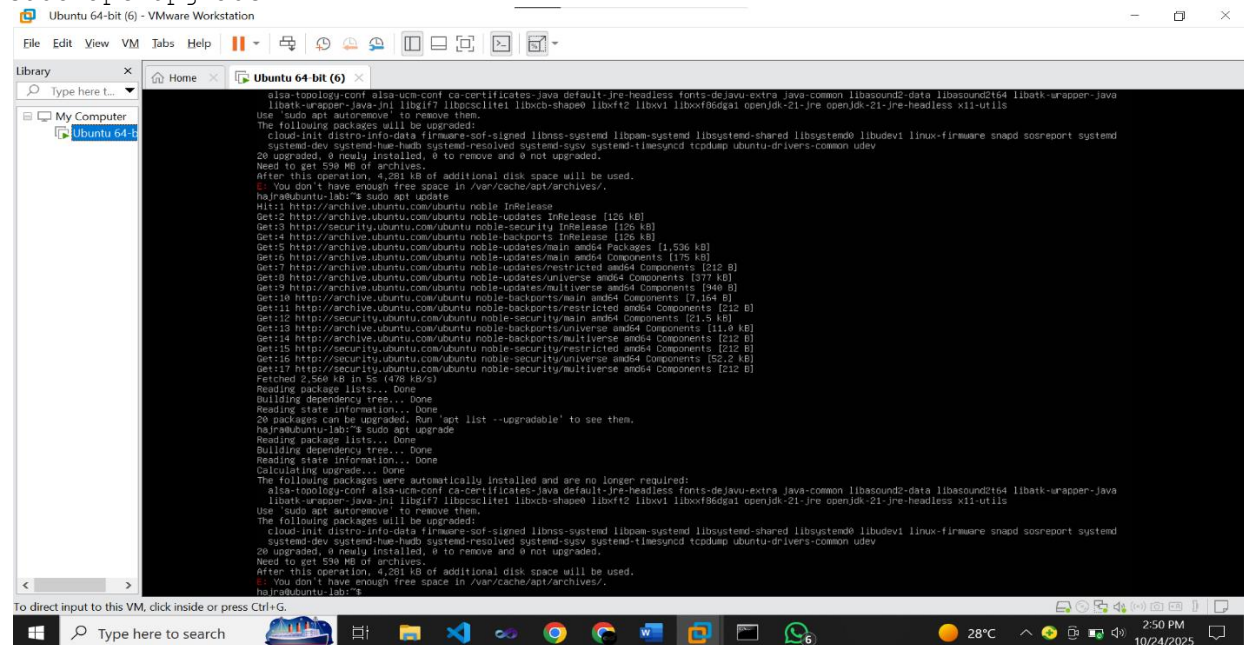
sudo apt update

```
hajra@ubuntu-lab:~$ sudo apt update
[sudo] password for hajra:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
20 packages can be upgraded. Run 'apt list --upgradable' to see them.
hajra@ubuntu-lab:~$
```

2. Upgrade installed packages

(This installs available updates for currently installed packages.)

sudo apt upgrade



Issue resolve steps:

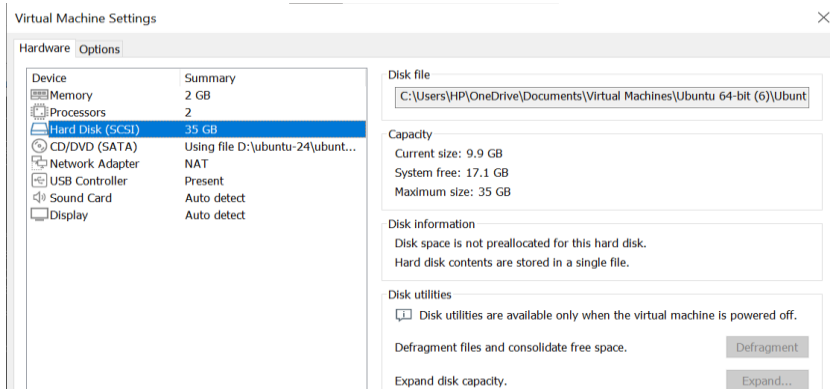
Step 1: Shut down Ubuntu

In your VM window:

```
sudo shutdown now
```

Step 2: Increase the disk size in VMware

1. In **VMware Workstation**, select your Ubuntu VM (don't power it on yet).
2. Go to **VM → Settings → Hard Disk (SCSI) → Utilities → Expand**
3. Increase it from (probably 20 GB) to **30 GB or 40 GB** → click **Expand**



25→35 GB expand

4. Click **OK** to save.

Step 3: Boot Ubuntu

1. Start your Ubuntu VM again.
2. Once you're logged in, open the terminal and run this:

```
df -h
```

```
hajra@ubuntu-lab:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            192M  1.6M  191M   1% /run
/dev/mapper/ubun-vg-ubuntu--lv 12G  11G  303M  98% /
tmpfs            960M    0  960M   0% /dev/shm
tmpfs            5.0M    0  5.0M   0% /run/lock
/dev/sda2        2.0G  192M  1.6G  11% /boot
tmpfs            192M  20K  192M   1% /run/user/1000
shm              64M    0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/483fff25e3b712de4439962c13861cbd9b245ecd1cc07a3a337a95
328491b2a/shm    64M    0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/15a028bbbf29088005fa373e0f59796cbf11b1c57575fa448e3e
shm              64M    0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/183356de05a5db846eb9baf5c6f530418c6a565183b35b0422032a
```

```
lsblk
```

```
hajra@ubuntu-lab:~$ sudo lsblk
[sudo] password for hajra:
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0        7:0    0   52.2M  1 loop /snap/aws-cli/1668
loop1        7:1    0   52.2M  1 loop /snap/aws-cli/1672
loop2        7:2    0   55.5M  1 loop /snap/core18/2952
loop3        7:3    0   63.8M  1 loop /snap/core20/2669
loop4        7:4    0   73.9M  1 loop /snap/core22/2133
loop5        7:5    0   66.8M  1 loop /snap/core24/1196
loop6        7:6    0   73.9M  1 loop /snap/core22/2139
loop7        7:7    0  140.6M  1 loop /snap/docker/3265
loop8        7:8    0   66.8M  1 loop /snap/core24/1225
loop9        7:9    0   44.1M  1 loop /snap/doctl/2263
loop10       7:10   0  420.3M  1 loop /snap/google-cloud-sdk/608
loop11       7:11   0   44.1M  1 loop /snap/doctl/2275
loop12       7:12   0  164.3M  1 loop /snap/microk8s/8511
loop13       7:13   0  118.3M  1 loop /snap/lxd/35624
loop14       7:14   0   22.7M  1 loop /snap/keepalived/3093
loop15       7:15   0  118.4M  1 loop /snap/lxd/36153
loop16       7:16   0   95.7M  1 loop /snap/kata-containers/2446
loop17       7:17   0  336M   1 loop /snap/nextcloud/50464
loop18       7:18   0   76.6M  1 loop /snap/powershell/313
loop19       7:19   0  164.2M  1 loop /snap/microk8s/8355
loop20       7:20   0    3.1M  1 loop /snap/mosquitto/1006
loop21       7:21   0   50.9M  1 loop /snap/snapd/25577
loop22       7:22   0   76.6M  1 loop /snap/powershell/316
loop23       7:23   0   13.9M  1 loop /snap/slcli/2957
loop24       7:24   0   50.8M  1 loop /snap/snapd/25202
loop25       7:25   0   23.2M  1 loop /snap/wormhole/509
sda          8:0    0    35G   0 disk
├─sda1        8:1    0     1M   0 part
├─sda2        8:2    0     2G   0 part /boot
└─sda3        8:3    0    23G   0 part
   └─ubuntu--vg-ubuntu--lv 252:0    0  11.5G   0 lvm  /
sr0          1:0    1  1024M   0 rom
```

Ubuntu uses **LVM (Logical Volume Manager)** – that's why main partition is:

```
/dev/mapper/ubun-vg-ubuntu--lv
```

So now we'll **extend** that logical volume to use the extra space you added (from 25 GB → 35 GB).

output tells us:

- Your main disk is /dev/sda (35 GB total)
- Your Linux partition is /dev/sda3 (currently 23 GB)
- Your logical volume /dev/ubuntu-vg/ubuntu-lv (root /) is only **11.5 GB** – that's the one we'll expand.

Grow the 3rd partition to fill the new disk space

```
sudo growpart /dev/sda 3
```

```
hajra@ubuntu-lab: $ sudo growpart /dev/sda 3
```

```
CHANGED: partition=3 start=4198400 old: size=48228352 end=52426751 new: size=69201887 end=73400286
```

This tells Ubuntu to extend partition 3 to use all available space on the disk.

Resize the physical volume

```
sudo pvresize /dev/sda3
```

```
hajra@ubuntu-lab: $ sudo pvresize /dev/sda3
```

```
Physical volume "/dev/sda3" changed
```

```
1 physical volume(s) resized or updated / 0 physical volume(s) not resized
```

This tells LVM that the underlying disk partition now has more space.

Extend your logical volume to use all free space

```
sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv
```

```
hajra@ubuntu-lab: $ sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv
```

```
Size of logical volume ubuntu-vg/ubuntu-lv changed from <11.50 GiB (2943 extents) to <33.00 GiB (8447 extents).
```

```
Logical volume ubuntu-vg/ubuntu-lv successfully resized.
```

This expands the main LVM volume (your root filesystem).

Finally, resize your filesystem

```
sudo resize2fs /dev/ubuntu-vg/ubuntu-lv
```

```
hajra@ubuntu-lab: $ sudo resize2fs /dev/ubuntu-vg/ubuntu-lv
```

```
resize2fs 1.47.0 (5-Feb-2023)
```

```
Filesystem at /dev/ubuntu-vg/ubuntu-lv is mounted on /; on-line resizing required
```

```
old_desc_blocks = 2, new_desc_blocks = 5
```

```
The filesystem on /dev/ubuntu-vg/ubuntu-lv is now 8649728 (4k) blocks long.
```

This makes your root filesystem actually use the new space.

After all that finishes successfully, confirm it with:

```
df -h
```

```
hajra@ubuntu-lab: $ df -h
Filesystem              Size  Used Avail Use% Mounted on
tmpfs                    192M  1.6M  191M   1% /run
/dev/mapper/ubuntu--vg-ubuntu-lv 33G  11G  21G  34% /
tmpfs                    960M   0  960M   0% /dev/shm
tmpfs                     5.0M   0   5.0M   0% /run/lock
/dev/sda2                 2.0G  192M  1.6G  11% /boot
tmpfs                    192M  20K  192M   1% /run/user/1000
shm                       64M   0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/483fff25e3b712de4439962c13861cbd9b245ecdcc07a3a337a95
128491b2a/shm
shm                       64M   0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/15a028bbbf29088005fa373e0f8e59796cbf11b1c57575fa448e3e
3d6f0da35/shm
shm                       64M   0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/183356de05a5db846eb9baf5c6f530418c6a505183b35b0422032a
d7fed3330/shm
```

You should now see / showing **around 33 GB total** instead of 12 GB.

SUMMARY:

What We Did

Ubuntu virtual machine (VM) was **running out of space** – the root partition (/) had only **12 GB total** and was **98% full**.

Even though i increased the virtual disk size in VMware (to 35 GB), Ubuntu didn't automatically use that extra space.

So, i had to **extend Ubuntu's internal partitions and filesystem** to use the new space.

Used growpart to Expand /dev/sda3

/dev/sda is my **main virtual hard disk**, and it's divided into smaller **partitions**:

Partition	Purpose	Size before	Notes
sda1	Boot loader space	1 MB	Used by GRUB to start Ubuntu
sda2	/boot partition	2 GB	Holds kernel and boot files
sda3	Main Linux partition (LVM)	23 GB → expanded to 33 GB	Holds almost all Ubuntu files

I expanded **only /dev/sda3** because that's where your **main Ubuntu data** lives.

Nothing changed in sda1 or sda2 – they stayed exactly the same.

I just used the *empty unallocated space* that was added after sda3 when the VMware disk was expanded.


```

hajra@ubuntu-lab:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  cloud-init distro-info-data firmware-sof-signed libnss-systemd libpam-systemd libsystemd-shared libsystemd0 libudev1 linux-firmware snapd sosreport systemd systemd-dev systemd-hwe-hwdb
  systemd-resolved systemd-sysv systemd-timesyncd tcpdump ubuntu-drivers-common udev
20 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 590 MB of archives.
After this operation, 4,281 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnss-systemd amd64 255.4-1ubuntu8.11 [159 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-dev all 255.4-1ubuntu8.11 [106 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-timesyncd amd64 255.4-1ubuntu8.11 [35.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-resolved amd64 255.4-1ubuntu8.11 [296 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd-shared amd64 255.4-1ubuntu8.11 [2,076 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd0 amd64 255.4-1ubuntu8.11 [434 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-sysv amd64 255.4-1ubuntu8.11 [11.9 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-systemd amd64 255.4-1ubuntu8.11 [235 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd amd64 255.4-1ubuntu8.11 [3,475 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 udev amd64 255.4-1ubuntu8.11 [1,874 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libudev1 amd64 255.4-1ubuntu8.11 [176 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ubuntu-drivers-common amd64 1:0.9.7.6ubuntu3.4 [65.8 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 distro-info-data all 0.60ubuntu0.5 [6,934 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-hwe-hwdb all 255.1.6 [3,684 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 tcpdump amd64 4.99.4-3ubuntu24.04.1 [479 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 firmware-sof-signed all 2023.12.1-1ubuntu1.10 [7,382 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-firmware amd64 20240318.git3b128b60-0ubuntu2.19 [538 MB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 snapd amd64 2.71+ubuntu24.04 [33.9 MB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 sosreport amd64 4.9.2-0ubuntu24.04.1 [372 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cloud-init all 25.2-0ubuntu1~24.04.1 [625 kB]
Fetched 590 MB in 4min 36s (2,135 kB/s)
Preconfiguring packages ...
(Reading database ... 142510 files and directories currently installed.)
Preparing to unpack .../0-libnss-systemd_255.4-1ubuntu8.11_amd64.deb ...
Unpacking libnss-systemd:amd64 (255.4-1ubuntu8.11) over (255.4-1ubuntu8.10) ...
Preparing to unpack .../1-systemd-dev_255.4-1ubuntu8.11_all.deb ...
Unpacking systemd-dev (255.4-1ubuntu8.11) over (255.4-1ubuntu8.10) ...
Preparing to unpack .../2-systemd-timesyncd_255.4-1ubuntu8.11_amd64.deb ...
Unpacking systemd-timesyncd (255.4-1ubuntu8.11) over (255.4-1ubuntu8.10) ...
Preparing to unpack .../3-systemd-resolved_255.4-1ubuntu8.11_amd64.deb ...
Unpacking systemd-resolved (255.4-1ubuntu8.11) over (255.4-1ubuntu8.10) ...
Preparing to unpack .../4-libsystemd-shared_255.4-1ubuntu8.11_amd64.deb ...
Unpacking libsystemd-shared:amd64 (255.4-1ubuntu8.11) over (255.4-1ubuntu8.10) ...
Setting up udev (255.4-1ubuntu8.11) ...
Setting up systemd-hwe-hwdb (255.1.6) ...
Setting up systemd-resolved (255.4-1ubuntu8.11) ...
Setting up snapd (2.71+ubuntu24.04) ...
Installing new version of config file /etc/apparmor.d/usr.lib.snapd.snap-confine.real ...
snapd.failure.service is a disabled or a static unit not running, not starting it.
snapd.gpio-chardev-setup.target is a disabled or a static unit not running, not starting it.
snapd.snap-repair.service is a disabled or a static unit not running, not starting it.
Setting up systemd-sysv (255.4-1ubuntu8.11) ...
Setting up ubuntu-drivers-common (1:0.9.7.6ubuntu3.4) ...
Setting up libnss-systemd:amd64 (255.4-1ubuntu8.11) ...
Setting up libpam-systemd:amd64 (255.4-1ubuntu8.11) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for rsyslog (8.2312.0-3ubuntu9.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for initramfs-tools (0.142ubuntu25.5) ...
update-initramfs: Generating /boot/initrd.img-6.8.0-86-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart faupd.service multipathd.service open-vm-tools.service polkit.service snap.keepalived.daemon.service ssh.service udisks2.service upower.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/needrestart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
hajra @ session #1: login[1532]
hajra @ session #3: apt[211349], sshd[11122]
hajra @ user manager service: systemd[2562]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
hajra@ubuntu-lab:~$

```

- Write a short explanation (3–5 sentences) describing the difference between apt update and apt upgrade:

nano ~/apt_update_vs_upgrade.md

```
hajra@ubuntu-lab:~$ nano ~/apt_update_vs_upgrade.md
```

```
hajra@ubuntu-lab:~$ hajra@ubuntu-lab:~$
```

Write your explanation, save, and exit Nano.

- Open the file or display it on screen:

```

hajra@ubuntu-lab:~$ nano ~/apt_update_vs_upgrade.md
hajra@ubuntu-lab:~$ cat ~/apt_update_vs_upgrade.md
The command 'sudo apt update' refreshes the list of available packages from online repositories. It does not install or remove anything; it just updates the local index.
On the other hand, 'sudo apt upgrade' actually installs the new versions of already-installed packages using the updated index.
Earlier, I faced issues where updates failed due to low disk space, and I had to extend the partition ('/dev/sda3') using 'growpart' to fix it. After fixing the partition, both apt update and upgrade ran successfully.
Without running apt update first, apt upgrade would not know about the latest versions available.
This process ensures the system stays up-to-date and secure.

```

Task 4: Install Visual Studio Code via Snap on CLI and Verify

Steps Performed:

- Installed Visual Studio Code using the snap command:

```
sudo snap install --classic code
```

```
hajra@ubuntu-lab:~$ sudo snap install --classic code
```

```
[sudo] password for hajra:
```

```
code 7d842fbb from Visual Studio Code (vscode) installed
```

- Verified that the package was successfully installed by listing it from snap:

```
snap list code
```

```
hajra@ubuntu-lab:~$ snap list code
Name Version Rev Tracking Publisher Notes
code 7d842fb8 211 latest/stable vscode classic
```

3. Checked the version of the installed application to confirm it was working correctly:

```
code --version
```

```
hajra@ubuntu-lab:~$ code --version
1.105.1
7d842fb85a0275a4a8e4d7e040d2625abbbf7f084
x64
```

or alternatively:

```
snap info code
```

```
hajra@ubuntu-lab:~$ snap info code
name: code
summary: Code editing. Redefined.
publisher: Visual Studio Code (vscode)
store-url: https://snapcraft.io/code
contact: https://twitter.com/code
license: unset
description: |
  Visual Studio Code is a new choice of tool that combines the
  simplicity of a code editor with what developers need for the core
  edit-build-debug cycle.
commands:
- code
- code.url-handler
snap-id: Ht0aUHi7ofh9Fbwh6m7jUN2pAy6kzBiu
tracking: latest/stable
refresh-date: today at 20:41 UTC
channels:
  latest/stable: 7d842fb8 2025-10-15 (211) 346MB classic
  latest/candidate: ↑
  latest/beta: ↑
  latest/edge: ↑
installed: 7d842fb8 (211) 346MB classic
```

4. Located where the snap placed the binary to ensure it was correctly linked in the system PATH:

```
ls -l /snap/bin | grep code
```

```
hajra@ubuntu-lab:~$ ls -l /snap/bin | grep code
lrwxrwxrwx 1 root root 13 Oct 24 20:41 code -> /usr/bin/snap
lrwxrwxrwx 1 root root 13 Oct 24 20:41 code.url-handler -> /usr/bin/snap
```

Task 5: Install XFCE GUI + XRDP and Launch VS Code

Steps Performed:

1. Connected to the virtual machine through SSH from the host terminal using:

```
ssh student@<vm-ip-address>
```

```
PS C:\Users\HP> ssh hajra@192.168.161.129
The authenticity of host '192.168.161.129 (192.168.161.129)' can't be established.
ED25519 key fingerprint is SHA256:M5unwK7+jxUhr6KM3bbMPRkghlGd+Kq5uspTQZ5Up3w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Warning: Permanently added '192.168.161.129' (ED25519) to the list of known hosts.
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-84-generic x86_64)
```

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:      https://ubuntu.com/pro
```

```
System information as of Sat 27 Sep 10:28:45 UTC 2025
```

```
System load: 1.46          Processes: 276
Usage of /: 79.2% of 11.21GB Users logged in: 1
Memory usage: 57%          IPv4 address for ens33: 192.168.161.129
Swap usage: 9%
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
0 updates can be applied immediately.
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

```
hajra@ubuntu-lab:~$
```

2. Updated the package lists and applied all available upgrades:

```
sudo apt update && sudo apt upgrade -y
```

```
hajra@ubuntu-lab:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for hajra:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

3. Installed the XFCE desktop environment and additional XFCE tools for a lightweight GUI:

```
sudo apt install xfce4 xfce4-goodies -y
```

```
hajra@ubuntu-lab: ~  
aspell-autobuildhash: processing: en [en-variant_1].  
aspell-autobuildhash: processing: en [en-variant_2].  
aspell-autobuildhash: processing: en [en-w_accents-only].  
aspell-autobuildhash: processing: en [en-wo_accents-only].  
aspell-autobuildhash: processing: en [en_AU-variant_0].  
aspell-autobuildhash: processing: en [en_AU-variant_1].  
aspell-autobuildhash: processing: en [en_AU-w_accents-only].  
aspell-autobuildhash: processing: en [en_AU-wo_accents-only].  
aspell-autobuildhash: processing: en [en_CA-variant_0].  
aspell-autobuildhash: processing: en [en_CA-variant_1].  
aspell-autobuildhash: processing: en [en_CA-w_accents-only].  
aspell-autobuildhash: processing: en [en_CA-wo_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ise-w_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ise-wo_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ize-w_accents-only].  
aspell-autobuildhash: processing: en [en_GB-ize-wo_accents-only].  
aspell-autobuildhash: processing: en [en_GB-variant_0].  
aspell-autobuildhash: processing: en [en_GB-variant_1].  
aspell-autobuildhash: processing: en [en_US-w_accents-only].  
aspell-autobuildhash: processing: en [en_US-wo_accents-only].  
Processing triggers for sgml-base (1.31) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
Running kernel seems to be up-to-date.  
Restarting services...  
Service restarts being deferred:  
/etc/needrestart/restart.d/dbus.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
No containers need to be restarted.  
User sessions running outdated binaries:  
hajra @ session #1: login[1532]  
hajra @ session #3: sshd[11122]  
hajra @ user manager service: systemd[2562]  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
hajra@ubuntu-lab: $
```

4. Installed and enabled the XRDP service to allow remote desktop access:

```
sudo apt install xrdp -y
```

```
hajra@ubuntu-lab: $ sudo apt install xrdp -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  libfuse2t64 libpipewire-0.3-modules-xrdp pipewire-module-xrdp xorgxrdp  
Suggested packages:  
  pipewire-bin guacamole  
The following NEW packages will be installed:  
  libfuse2t64 libpipewire-0.3-modules-xrdp pipewire-module-xrdp xorgxrdp xrdp  
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.  
Service restarts being deferred:  
/etc/needrestart/restart.d/dbus.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
No containers need to be restarted.  
User sessions running outdated binaries:  
hajra @ session #1: login[1532]  
hajra @ session #3: sshd[11122]  
hajra @ user manager service: systemd[2562]  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
hajra@ubuntu-lab: $  
sudo systemctl enable --now xrdp  
hajra@ubuntu-lab: $ sudo systemctl enable --now xrdp  
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp
```

5. Verified that the XRDP service was running properly:

```
sudo systemctl status xrdp
```

```
hajra@ubuntu-lab: $ sudo systemctl status xrdp  
● xrdp.service - xrdp daemon  
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)  
   Active: active (running) since Fri 2025-10-24 21:03:34 UTC; 3min 27s ago  
     Docs: man:xrdp(8)  
           man:xrdp.ini(5)  
   Main PID: 464810 (xrdp)  
     Tasks: 1 (limit: 2210)  
    Memory: 936.0K (peak: 1.6M)  
       CPU: 72ms  
    CGroup: /system.slice/xrdp.service  
            └─464810 /usr/sbin/xrdp  
  
Oct 24 21:03:33 ubuntu-lab systemd[1]: Starting xrdp.service - xrdp daemon...  
Oct 24 21:03:33 ubuntu-lab xrdp[464808]: [INFO ] address [0.0.0.0] port [3389] mode 1  
Oct 24 21:03:33 ubuntu-lab xrdp[464808]: [INFO ] listening to port 3389 on 0.0.0.0  
Oct 24 21:03:33 ubuntu-lab xrdp[464808]: [INFO ] xrdp_listen_pp done  
Oct 24 21:03:33 ubuntu-lab systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after start: No such file or directory  
Oct 24 21:03:34 ubuntu-lab systemd[1]: Started xrdp.service - xrdp daemon.  
Oct 24 21:03:35 ubuntu-lab xrdp[464810]: [INFO ] starting xrdp with pid 464810  
Oct 24 21:03:35 ubuntu-lab xrdp[464810]: [INFO ] address [0.0.0.0] port [3389] mode 1  
Oct 24 21:03:35 ubuntu-lab xrdp[464810]: [INFO ] listening to port 3389 on 0.0.0.0  
Oct 24 21:03:35 ubuntu-lab xrdp[464810]: [INFO ] xrdp_listen_pp done
```

6. Configured the XRDP session to use XFCE as the default desktop environment:

```
echo xfce4-session > ~/.xsession
```

```
hajra@ubuntu-lab:~$ echo xfce4-session > ~/.xsession  
hajra@ubuntu-lab:~$
```

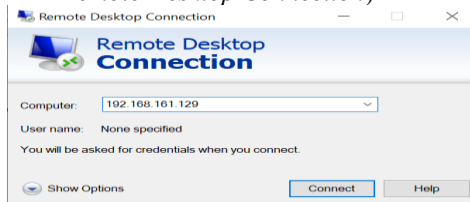
7. From the host system, connected to the VM using Remote Desktop Connection (mstsc) and logged in using the Ubuntu username and password.

My VM's IP is 192.168.161.129

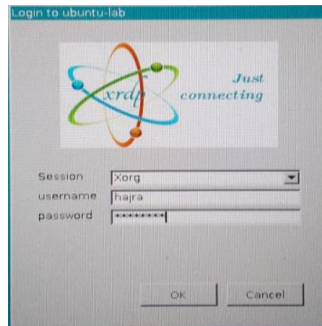
Now do this on Windows host system (not inside PowerShell):

1. Press Windows + R
2. Type mstsc and press Enter

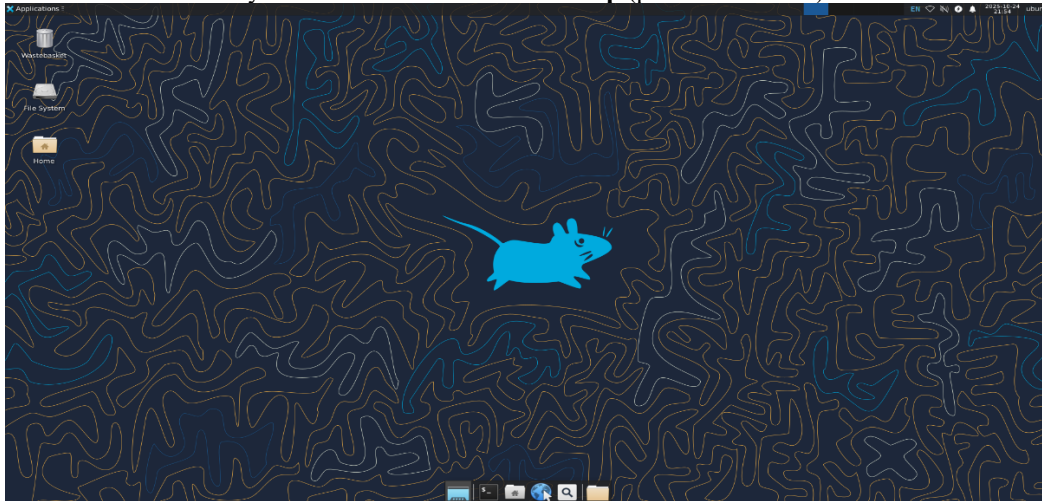
3. In the window that opens (titled *Remote Desktop Connection*):



4. When prompted for credentials:

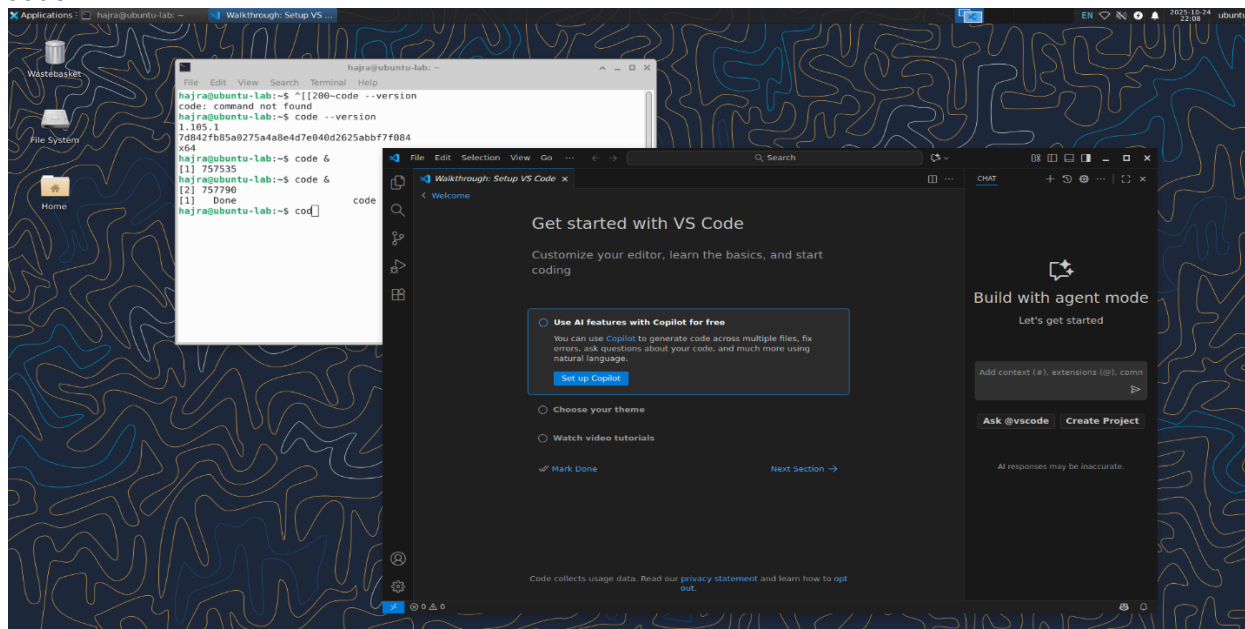


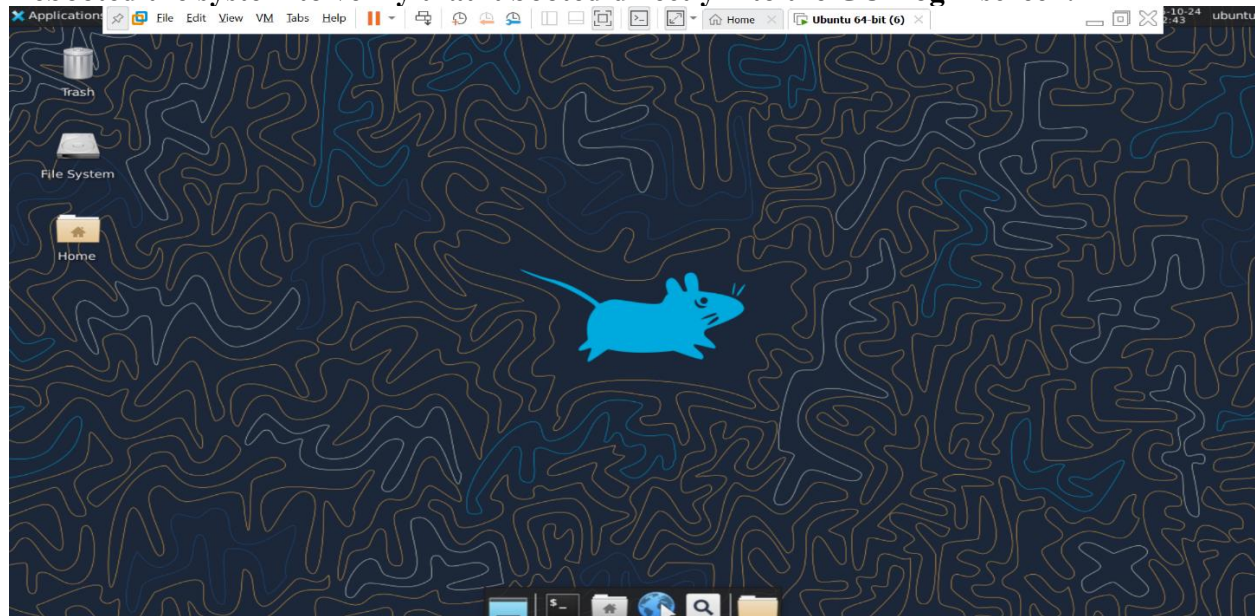
5. If a certificate warning appears → click **Yes / Continue**
6. Wait a few seconds — you should see the **XFCE desktop** (panel and menus).



8. Once logged into the GUI, launched Visual Studio Code to verify that the graphical interface and installed software worked properly:

Code





7. **Disabled GUI login at boot (to return to text-based CLI mode) by switching to the multi-user target:**

```
sudo systemctl set-default multi-user.target
sudo systemctl disable lightdm
```

```
hajra@ubuntu-lab:~$ sudo systemctl set-default multi-user.target
[sudo] password for hajra:
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
hajra@ubuntu-lab:~$ sudo systemctl disable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable lightdm
Removed "/etc/systemd/system/display-manager.service".
hajra@ubuntu-lab:~$
```

8. **Rebooted again to confirm that the system now starts in CLI mode.**

```
Ubuntu 24.04.3 LTS ubuntu-lab tty1
ubuntu-lab login: hajra
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed 22 Oct 12:09:53 UTC 2025

System load: 0.96          Processes: 270
Usage of /: 86.9% of 11.21GB Users logged in: 1
Memory usage: 57%         IPv4 address for ens33: 192.168.161.129
Swap usage: 9%

=> / is using 86.9% of 11.21GB

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

hajra@ubuntu-lab:~$
```

9. **Tested manual GUI control (starting and stopping the GUI without reboot):**

```
sudo systemctl start lightdm
```

```
hajra@ubuntu-lab:~$ sudo systemctl start lightdm
[sudo] password for hajra:
```

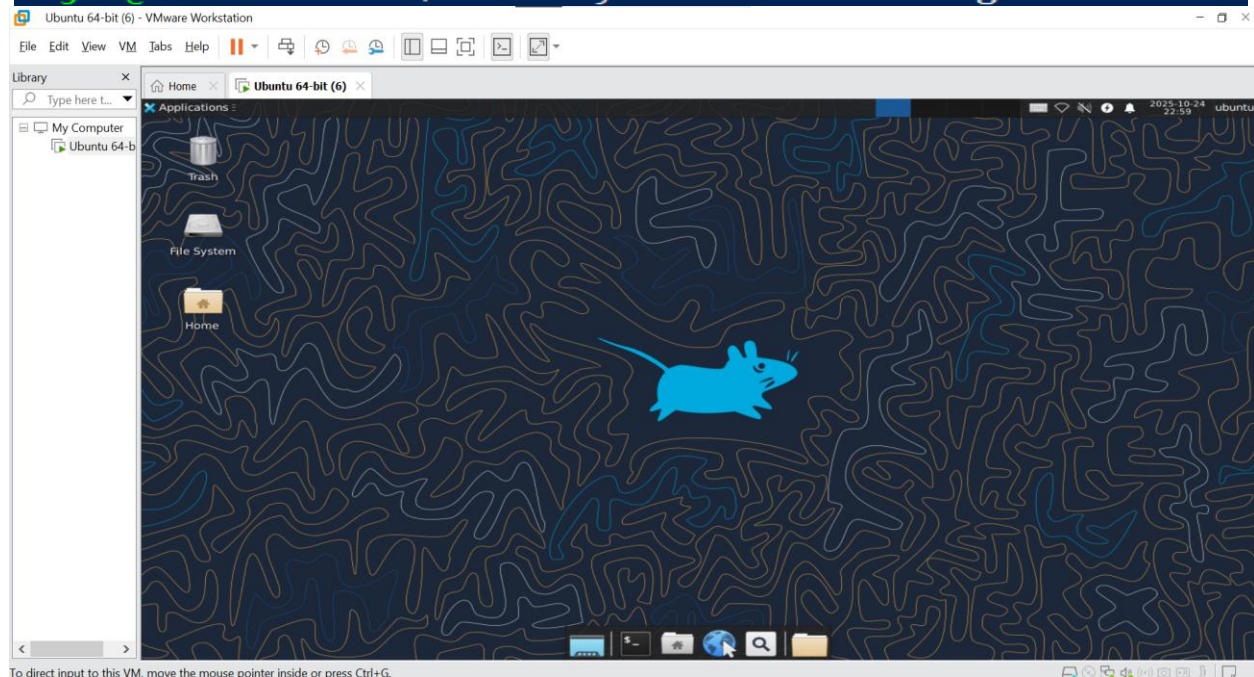
```
sudo systemctl stop lightdm
```

```
hajra@ubuntu-lab:~$ sudo systemctl stop lightdm
hajra@ubuntu-lab:~$
```

10. **Started the GUI session manually when in CLI mode:**

```
sudo systemctl start lightdm
```

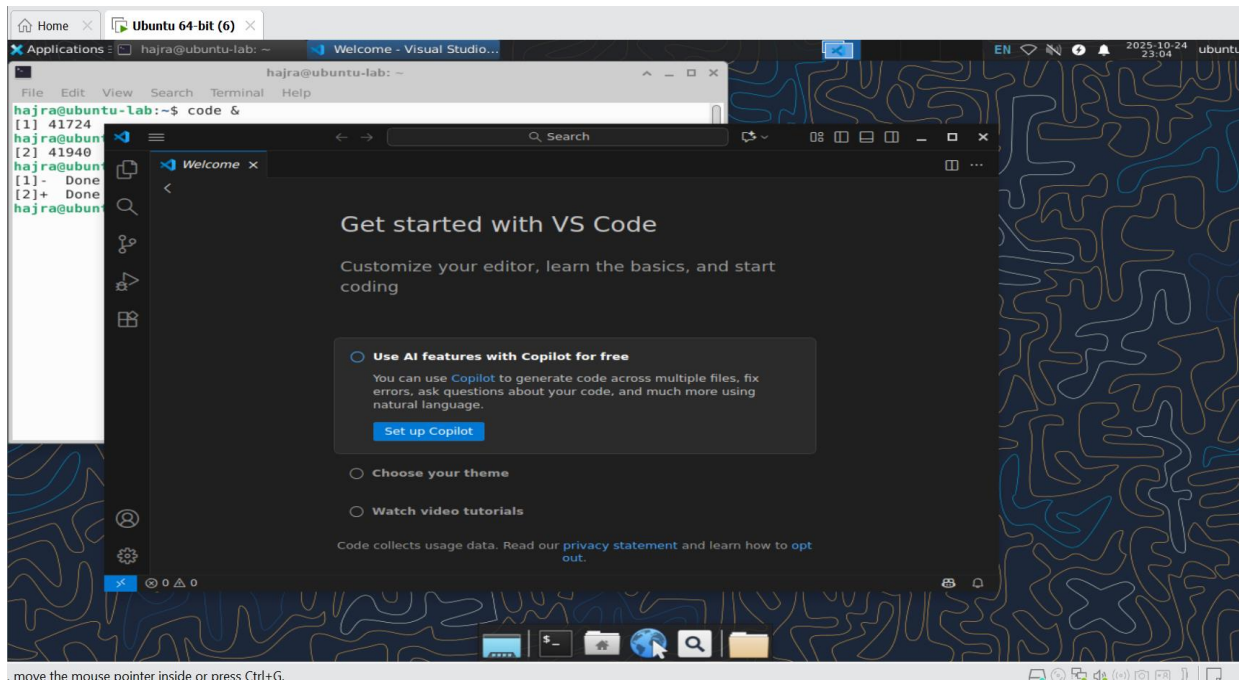
```
hajra@ubuntu-lab:~$ sudo systemctl start lightdm
```



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

11. **Inside the GUI environment, launched Visual Studio Code (installed earlier in Task 4) to verify that it runs properly:**

Code



Task 7: Install Google Chrome by Adding Its APT Source and Key

Steps Performed:

1. Attempted to install Google Chrome directly before adding its repository and key to observe the expected failure:

```
sudo apt install google-chrome-stable -y
hajra@ubuntu-lab:~$ sudo apt install google-chrome-stable -y
[sudo] password for hajra:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-chrome-stable
```

2. Inspected the APT configuration to understand why the installation failed by listing the `/etc/apt` directory:

```
ls -la /etc/apt
```

```
hajra@ubuntu-lab:~$ ls -la /etc/apt
total 48
drwxr-xr-x  9 root root 4096 Sep 26 21:06 .
drwxr-xr-x 147 root root 12288 Oct 24 21:25 ..
drwxr-xr-x  2 root root 4096 Oct 24 20:19 apt.conf.d
drwxr-xr-x  2 root root 4096 Mar 31 2024 auth.conf.d
drwxr-xr-x  2 root root 4096 Mar 31 2024 keyrings
drwxr-xr-x  2 root root 4096 Aug  5 17:14 preferences.d
drwxr-xr-x  2 root root 4096 Aug  5 17:14 preferences.d.save
-rw-r--r--  1 root root  70 Sep 26 21:24 sources.list
drwxr-xr-x  2 root root 4096 Sep 26 21:24 sources.list.d
drwxr-xr-x  2 root root 4096 Aug  5 17:01 trusted.gpg.d
```

3. Viewed the main system sources file to confirm that Google's repository was not listed:

```
cat /etc/apt/sources.list
hajra@ubuntu-lab:~$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
```

4. Checked additional repository files in `/etc/apt/sources.list.d/`:

```
ls -la /etc/apt/sources.list.d/
hajra@ubuntu-lab:~$ ls -la /etc/apt/sources.list.d/
total 16
drwxr-xr-x  2 root root 4096 Sep 26 21:24 .
drwxr-xr-x  9 root root 4096 Sep 26 21:06 ..
-rw-r--r--  1 root root 382 Sep 26 21:06 ubuntu.sources
-rw-r--r--  1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
```

5. Opened or created the `ubuntu.sources` file to verify that Google's Chrome repository was missing:

```
cat /etc/apt/sources.list.d/ubuntu.sources
```

```

hajra@ubuntu-lab: $ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

```

6. Added Chrome's repository information manually to the sources file:

```
sudo nano /etc/apt/sources.list.d/ubuntu.sources
```

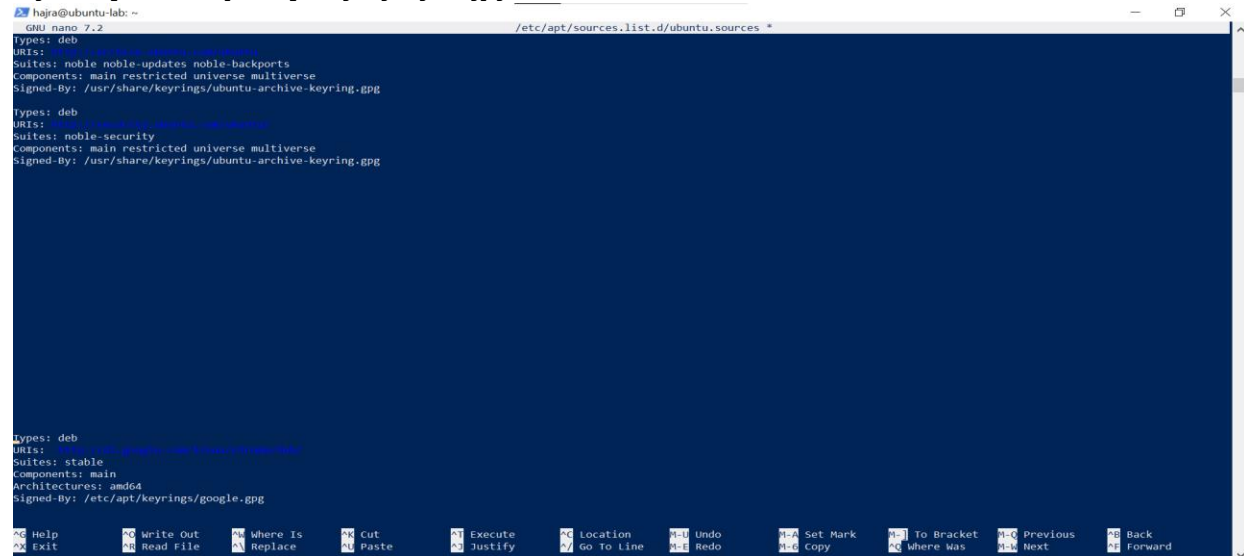
```
hajra@ubuntu-lab: $ sudo nano /etc/apt/sources.list.d/ubuntu.sources
```

Then appended:

```

Types: deb
URIs: http://dl.google.com/linux/chrome/deb/
Suites: stable
Components: main
Architectures: amd64
Signed-By: /etc/apt/keyrings/google.gpg

```



7. Created the keyrings directory (if missing) and imported Google's official signing key:

```
curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
```

```
hajra@ubuntu-lab: $ sudo mkdir -p /etc/apt/keyrings
```

```
hajra@ubuntu-lab: $ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
```

```
hajra@ubuntu-lab: $
```

8. Updated the system package index and successfully installed Chrome:

```
sudo apt update
```

```

hajra@ubuntu-lab: $ sudo apt update
Get:1 http://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,217 B]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,541 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [292 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,170 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [489 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,128 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 5,521 kB in 9s (608 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.

```

```
sudo apt install google-chrome-stable -y
```

```

hajra@ubuntu-lab: $ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fonts-liberation fonts-liberation-sans-narrow
The following NEW packages will be installed:
  fonts-liberation fonts-liberation-sans-narrow google-chrome-stable
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 122 MB of archives.
After this operation, 401 MB of additional disk space will be used.
Get:1 http://dl.google.com/linux/chrome/deb stable/main amd64 google-chrome-stable amd64 141.0.7390.122-1 [120 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-liberation all 1:2.1.5-3 [1,603 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-liberation-sans-narrow all 1:1.07.6-4 [197 kB]
Fetched 122 MB in 1min 11s (1,725 kB/s)

```



```
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
hajra@ubuntu-lab: $
```

9. (Alternate – Preferred Method)

Removed the earlier Chrome installation and cleaned up the previous configuration:

```
sudo apt remove google-chrome-stable -y
```

```
hajra@ubuntu-lab: $ sudo apt remove google-chrome-stable -y
[sudo] password for hajra:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  google-chrome-stable
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 396 MB disk space will be freed.
(Reading database ... 183468 files and directories currently installed.)
Removing google-chrome-stable (141.0.7390.122-1) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
```

```
sudo nano /etc/apt/sources.list.d/ubuntu.sources # removed chrome stanza
```

```
hajra@ubuntu-lab:~$ sudo nano /etc/apt/sources.list.d/ubuntu.sources
```

```
hajra@ubuntu-lab: ~
GNU nano 7.2 /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```
sudo rm -f /etc/apt/keyrings/google.gpg
```

```
hajra@ubuntu-lab:~$ sudo rm -f /etc/apt/keyrings/google.gpg
```

```
hajra@ubuntu-lab:~$
```

10. Created a dedicated one-line Chrome repository file for a cleaner configuration:

```
echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/google.gpg]
http://dl.google.com/linux/chrome/deb/ stable main" | sudo tee
/etc/apt/sources.list.d/google-chrome.list > /dev/null
```

```
hajra@ubuntu-lab:~$ echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/google.gpg] http://dl.google.com/linux/chrome/deb/ stable main" | sudo tee /etc/apt/sources.list.d/google-chrome.list > /dev/null
hajra@ubuntu-lab:~$
```

11. Verified the new sources list file:

```
ls -la /etc/apt/sources.list.d/
```

```
hajra@ubuntu-lab:~$ ls -la /etc/apt/sources.list.d/
total 20
drwxr-xr-x 2 root root 4096 Oct 25 04:41 .
drwxr-xr-x 9 root root 4096 Sep 26 21:06 ..
-rw-r--r-- 1 root root 107 Oct 25 04:45 google-chrome.list
-rw-r--r-- 1 root root 384 Oct 25 04:41 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
```

12. Re-added Google's signing key using the keyrings method:

```
sudo mkdir -p /etc/apt/keyrings
```

```
curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
```

```
hajra@ubuntu-lab:~$ sudo mkdir -p /etc/apt/keyrings
```

```
hajra@ubuntu-lab:~$ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
hajra@ubuntu-lab:~$
```

13. Updated the package list again and reinstalled Google Chrome successfully using the new method:

```
sudo apt update
```

```
hajra@ubuntu-lab:~$ sudo apt update
Hit:1 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

```

sudo apt install google-chrome-stable -y
hajra@ubuntu-lab:~$ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  google-chrome-stable
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 120 MB of archives.
After this operation, 396 MB of additional disk space will be used.
Get:1 http://dl.google.com/linux/chrome/deb stable/main amd64 google-chrome-stable amd64 141.0.7390.122-1 [120 MB]
Fetched 120 MB in 1min 1s (1,960 kB/s)
Selecting previously unselected package google-chrome-stable.
(Reading database ... 18198 files and directories currently installed.)
Preparing to unpack .../google-chrome-stable_141.0.7390.122-1_amd64.deb ...
Unpacking google-chrome-stable (141.0.7390.122-1) ...
Setting up google-chrome-stable (141.0.7390.122-1) ...
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/x-www-browser (x-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/gnome-www-browser (gnome-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/google-chrome (google-chrome) in auto mode
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
hajra@ubuntu-lab:~$

```

Task 8: Install Applications via PPA (Audacity & OBS Studio) and Launch Steps Performed:

1. Added the Audacity PPA to the system and updated the package index:

```
sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y
```

```

hajra@ubuntu-lab:~$ sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu/
Suites: noble
Components: main
Description:
  Unofficial build of Audacity audio editor
For help, please use Audacity forum: http://forum.audacityteam.org/
If the packages here are helpful, you may buy me a coffee:
  https://ko-fi.com/ubuntuhandbook1
More info: https://launchpad.net/~ubuntuhandbook1/+archive/ubuntu/audacity
Adding repository.
Hit:1 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease [18.1 kB]
Get:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main amd64 Packages [1,064 B]
Get:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main Translation-en [492 B]
Fetched 19.6 kB in 3s (5,935 B/s)
Reading package lists... Done

```

```
sudo apt update
```

```

hajra@ubuntu-lab:~$ sudo apt update
Hit:1 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.

```

2. Installed Audacity using the apt command:

```
sudo apt install audacity -y
```

```

hajra@ubuntu-lab:~$ sudo apt install audacity -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  audacity-data libdouble-conversion3 libflac++10 libid3tag0 libilv-0-0 libmd4c0 libopusfile0 libpcre2-16-0 libpcre2-32-0 libportaudio2 libportmidi0 libportsmf0t64 libqt5core5t64
  libqt5dbus5t64 libqt5gui5t64 libqt5network5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5svg5 libqt5waylandclient5 libqt5waylandcompositor5 libqt5widgets5t64 libqt5xml5t64
  libqt5xmlpatterns5 libsndfile0 libsoxr0 libsoxr0-0 libsoundtouch1 libsratom-0-0 libswi-0-0 libvamp-hostsdk3v5 libwxbase3.2-1t64 libwxgtk3.2-1t64 libxcb-icccm4 libxcb-image0 libxcb-keyyms1
  libxcb-render-util0 libxcb-xinerama0 libxcb-xinput0 libxcb-xxkb1 libxcbxxcommon-x11-0 libzix-0-0 qt5-gtk-platformtheme qttranslations5-l10n qtwayland5
Suggested packages:
  ladspa-plugin qgnomeplatform-qt5 qt5-image-formats-plugins qt5-qmltooling-plugins serdi sordi
The following NEW packages will be installed:
  audacity audacity-data libdouble-conversion3 libflac++10 libid3tag0 libilv-0-0 libmd4c0 libopusfile0 libpcre2-16-0 libpcre2-32-0 libportaudio2 libportmidi0 libportsmf0t64
  libqt5core5t64 libqt5dbus5t64 libqt5gui5t64 libqt5network5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5svg5 libqt5waylandclient5 libqt5waylandcompositor5 libqt5widgets5t64
  libqt5xml5t64 libqt5xmlpatterns5 libsndfile0 libsoxr0 libsoxr0-0 libsoundtouch1 libsratom-0-0 libswi-0-0 libvamp-hostsdk3v5 libwxbase3.2-1t64 libwxgtk3.2-1t64 libxcb-icccm4 libxcb-image0
  libxcb-keyyms1 libxcb-render-util0 libxcb-xinerama0 libxcb-xinput0 libxcb-xxkb1 libxcbxxcommon-x11-0 libzix-0-0 qt5-gtk-platformtheme qttranslations5-l10n qtwayland5
0 upgraded, 46 newly installed, 0 to remove and 0 not upgraded.
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
hajra@ubuntu-lab:~$

```

3. **Verified or launched Audacity** from the terminal or GUI to confirm successful installation:

```
audacity --version
audacity
```

```
hajra@ubuntu-lab:~$ which audacity
/usr/bin/audacity
hajra@ubuntu-lab:~$ dpkg -l | grep audacity
ii  audacity          3.7.5-0build1~ubuntu24.04      amd64      fast, cross-platform audio editor
ii  audacity-data     3.7.5-0build1~ubuntu24.04      all        fast, cross-platform audio editor (data)
```

4. **Added the OBS Studio PPA and updated the package list again:**

```
sudo add-apt-repository ppa:obsproject/obs-studio -y
```

```
hajra@ubuntu-lab:~$ sudo add-apt-repository ppa:obsproject/obs-studio -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu/
Suites: noble
Components: main
'
Description:
Latest stable release of OBS Studio
More info: https://launchpad.net/~obsproject/+archive/ubuntu/obs-studio
Adding repository.
Hit:1 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Get:5 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease [17.8 kB]
Hit:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:8 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main amd64 Packages [1,172 B]
Get:9 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main Translation-en [160 B]
Fetched 19.1 kB in 5s (3,947 B/s)
Reading package lists... Done
```

```
sudo apt update
```

```
hajra@ubuntu-lab:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:6 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

5. **Installed OBS Studio using apt:**

```
sudo apt install obs-studio -y
```

```
hajra@ubuntu-lab:~$ sudo apt install obs-studio -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
libbaes0 libb69 libbdevice0 libbfilter9 libbformat6 libbz2-1 libbdlplus libblas3 libbluray2 libbs2b0 libchromaprint1 libjson1 libdc1394-25 libdecor-0 libdecor-0-plugin-1-gtk
libfdk-aac2 libfftw3-double3 libflite1 libgfortran5 libgme0 liblapack3 liblua5.1-2 liblua5.1-common libmbedcrypto764 libmbdts14t64 libmbdx509-1t64 libmysofa1 libnormt64
libopenal-data libopenal1 libopenmpt0t64 libpgm-5.3-0t64 libplacebo338 libpocketsphinx3 libpostproc57 libqcodegencpp1 libqt6core6t64 libqt6dbus6t64 libqt6gui6t64 libqt6network6t64
libqt6opengl6t64 libqt6qml6 libqt6qmlmodels6 libqt6quick6 libqt6svg6 libqt6waylandclient6 libqt6waylandcompositor6 libqt6waylandeglclient6 libqt6waylandintegration6
libqt6waylandintegration6 libqt6widgets6t64 libqt6xml6t64 librabbitmq4 librist4 librubberband2 libsd12-2.0-0 libsndio7.0 libspatialbase3t64
libstr1.5-gnutls libstr1.5-openssl libssh-gcrypt-4 libswscale7 libt6t64 libudfread0 libunibreak5 libvidstab1.1 libxcb-composite0 libzim2 libzmq5 pocketsphinx-en-us
qt6-gtk-platformtheme qt6-qt5-plugins qt6-translations-110n qt6-wayland
Suggested packages:
libbluray-bdj libfftw3-bin libfftw3-dev qt6-qmltooling-plugins sndiod
The following NEW packages will be installed:
libbaes0 libb69 libbdevice0 libbfilter9 libbformat6 libbz2-1 libbdlplus libblas3 libbluray2 libbs2b0 libchromaprint1 libjson1 libdc1394-25 libdecor-0 libdecor-0-plugin-1-gtk
libfdk-aac2 libfftw3-double3 libflite1 libgfortran5 libgme0 liblapack3 liblua5.1-2 liblua5.1-common libmbedcrypto764 libmbdts14t64 libmbdx509-1t64 libmysofa1 libnormt64
libopenal-data libopenal1 libopenmpt0t64 libpgm-5.3-0t64 libplacebo338 libpocketsphinx3 libpostproc57 libqcodegencpp1 libqt6core6t64 libqt6dbus6t64 libqt6gui6t64 libqt6network6t64
libqt6opengl6t64 libqt6qml6 libqt6qmlmodels6 libqt6quick6 libqt6svg6 libqt6waylandclient6 libqt6waylandcompositor6 libqt6waylandeglclient6 libqt6waylandintegration6
libqt6waylandintegration6 libqt6widgets6t64 libqt6xml6t64 librabbitmq4 librist4 librubberband2 libsd12-2.0-0 libsndio7.0 libspatialbase3t64
libstr1.5-gnutls libstr1.5-openssl libssh-gcrypt-4 libswscale7 libt6t64 libudfread0 libunibreak5 libvidstab1.1 libxcb-composite0 libzim2 libzmq5 obs-studio pocketsphinx-en-us
qt6-gtk-platformtheme qt6-qt5-plugins qt6-translations-110n qt6-wayland
0 upgraded, 75 newly installed, 0 to remove and 0 not upgraded.
```

6. **Verified or launched OBS Studio** to ensure that the installation completed successfully:

```
obs --version
```

```
hajra@ubuntu-lab:~$ obs --version
OBS Studio - 32.0.0
```

Task 9: Create a Kubernetes Sample YAML Using Vim

Steps Performed:

1. **Checked if vim was installed:**

```
Vim
```

```
~$ Select hajra@ubuntu-lab: ~
```

```
VIM - Vi IMproved
version 9.1.697
by Bram Moolenaar et al.
Modified by teamvim@tracker.debian.org
Vim is open source and freely distributable

Sponsor Vim development!
type :help sponsor for information
type :q to exit
type :help or :? for on-line help
type :help version9 for version info
```

for just checking vim (no changes), you can do:
:q!

2. **Created the Lab5 working directory and navigated into it:**

```
mkdir -p ~/Lab5
cd ~/Lab5
pwd
```

```
hajra@ubuntu-lab:~$ mkdir -p ~/Lab5
hajra@ubuntu-lab:~$ cd ~/Lab5
hajra@ubuntu-lab:~/Lab5$ pwd
/home/hajra/Lab5
```

3. **Created the Kubernetes sample YAML file using vim:**

```
vim k8s-sample.yaml
```

```
hajra@ubuntu-lab:~/Lab5$ vim k8s-sample.yaml
```

- o Enabled insert mode (i) and pasted the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
  - name: nginx
    image: nginx:1.19
    ports:
    - containerPort: 80
  restartPolicy: Always
```



4. **Saved and exited vim:**

- o Pressed Esc and then typed :wq to save the file.

```
hajra@ubuntu-lab:~/Lab5$ vim k8s-sample.yaml
hajra@ubuntu-lab:~/Lab5$
```

5. **Verified that the file was saved correctly:**

```
ls -la
```

```
hajra@ubuntu-lab:~/Lab5$ ls -la
total 12
drwxrwxr-x  2 hajra hajra 4096 Oct 25 07:43 .
drwxr-x--- 23 hajra hajra 4096 Oct 25 07:43 ..
-rw-rw-r--  1 hajra hajra  320 Oct 25 07:43 k8s-sample.yaml
```

Task 10: Edit the Kubernetes YAML: Add Annotation and Discard Temporary Change

Steps Performed:

1. **Opened the manifest file in vim:**

```
cd ~/Lab5
```

```
vim k8s-sample.yaml
```

```
hajra@ubuntu-lab:~/Lab5$ cd ~/Lab5
```

```
hajra@ubuntu-lab:~/Lab5$ vim k8s-sample.yaml
```

- o Entered **insert mode** by pressing i.

2. **Added a permanent annotation** under the metadata section (matching YAML indentation):

```
annotations:
lab: lesson11
```



```
hajra@ubuntu-lab: ~/Lab5
```

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
```

- Pressed `Esc` and saved the file with `:wq`.

3. Verified the annotation was added correctly:

```
cat k8s-sample.yaml
```

```
hajra@ubuntu-lab:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

4. Practiced a temporary edit:

- Reopened the file:
`vim k8s-sample.yaml`

```
hajra@ubuntu-lab: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

- Entered insert mode (`i`) and added a temporary comment:
`# temp: do-not-keep`

```
hajra@ubuntu-lab: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
            # temp: do-not-keep
```

5. Discarded the temporary change without saving:

- Pressed `Esc` and exited with `:q!`.

```
hajra@ubuntu-lab:~/Lab5$ vim k8s-sample.yaml
hajra@ubuntu-lab:~/Lab5$
```

6. Verified that the temporary comment was not saved:

```
cat k8s-sample.yaml
```

```
hajra@ubuntu-lab:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

Task 11: Vim Editing Practice: Delete, Undo, Numeric Deletes, and Navigation

Steps Performed

1. Opened the Kubernetes YAML file in vim

```
cd ~/Lab5
```

```
vim k8s-sample.yaml
```

```
hajra@ubuntu-lab:~$ cd ~/Lab5
hajra@ubuntu-lab:~/Lab5$ vim k8s-sample.yaml
```

2. Deleted a single line using `dd` and then undone it

- o In **command mode**, navigated to the line containing:

```
image: nginx:1.19
```

```
- name  nginx
  image  nginx:1.19
  ports
```

- o Deleted that line using:

```
Dd
```

```
- name  nginx
  ports
```

- o Immediately **undid** the deletion with:

```
U
```

```
- name  nginx
  image  nginx:1.19
  ports
```

- o The file was restored to its original state.

3. Deleted three lines at once using numeric delete and undone it

```
apiVersion  v1
kind  Pod
metadata
```

- o In **command mode**, moved the cursor to the start of the section and ran:

```
3dd
```

```
annotations:
  lab_lesson11
  name  nginx-pod
  spec
containers
```

(used numeric prefix form `3dd`)

- o This deleted **three lines at once**.
- o Immediately undid the deletion with:

```
U
```

```
apiVersion  v1
kind  Pod
metadata
```

4. Navigation Practice

```
restartPolicy  Always
# temp: do-not-keep
```

- o Jumped to the **first line** of the file:

```
1G
```

```
apiVersion  v1
kind  Pod
metadata
```

(Screenshot: *task11_line1.png*)

- o Jumped to the **last line**:

```
G
```

```
restartPolicy  Always
# temp: do-not-keep
```

- o Practiced **line navigation** within a line:

```
restartPolicy  Always
# temp: do-not-keep
```

- Moved to the **end of the line**: `$`

```
# temp: do-not-keep
```

- Moved back to the **start of the line**: `0`

```
# temp: do-not-keep
```

5. Exited vim without saving any changes:

```
:q
```

```
hajra@ubuntu-lab:~/Lab5$ vim k8s-sample.yaml
hajra@ubuntu-lab:~/Lab5$
```

Task 12: Vim Search, Add Matches, Substitute, and Undo

Steps Performed

1. Opened the file in vim

```
cd ~/Lab5
```

```
vim k8s-sample.yaml
```

```
hajra@ubuntu-lab:~$ cd ~/Lab5
hajra@ubuntu-lab:~/Lab5$ vim k8s-sample.yaml
```

2. Searched for the string “nginx”

- o In **command mode**, typed:
/nginx
and pressed **Enter**.

```
restartPolicy Always
# temp: do-not-keep
```

```
name nginx-pod
```

- o The first match was **highlighted**, and the cursor moved to it.

3. Navigated between matches

- o Moved to the **next match** with:
N

```
lab lesson11
name nginx-pod
spec
  containers
  - name nginx
    image nginx:1.19
```

- o Moved back to the **previous match** with:
N

```
name nginx-pod
spec
  containers
  - name nginx
```

4. Added two more occurrences of “nginx”

- o Entered **insert mode** (i) and added two comment lines with the word **nginx** (for example, under the metadata section):
nginx test 1
nginx test 2
- o Pressed **Esc** and saved the file with:

```
:w
hajra@ubuntu-lab:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
    name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
            # temp: do-not-keep
            # nginx added for testing
            # # another nginx line
```

5. Verified cycling through all matches

- o Searched again for the same string:
/nginx

```
lab lesson11
name nginx-pod
```

- o Pressed **n** multiple times to cycle through each match and confirmed vim highlighted every occurrence in sequence.

```
image nginx:1.19
```

6. Substituted all “nginx” occurrences with “webapp”

- o In **command mode**, ran:
:%s/nginx/webapp

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
    name: webapp-pod
  spec:
    containers:
      - name: webapp
        image: webapp:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
            # temp: do-not-keep
            # webapp added for testing
            # # another nginx line
```

- This replaced every instance of **nginx** with **webapp** throughout the file.

7. Undid the substitution

- Pressed:

U

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    lab: lesson11
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
        # temp: do-not-keep
```

- This **reverted** the file back to its previous state (before the substitution).

8. Exited vim without saving unintended changes

:q!

```
hajra@ubuntu-lab: ~/Lab5$ vim k8s-sample.yaml
hajra@ubuntu-lab: ~/Lab5$
```

(Used if any temporary edits needed to be discarded.)

Exam Evaluation Question: Docker Desktop Installation

Steps Performed

1. Updated the system packages to make sure all repositories and dependencies were current:

`sudo apt update`

```
hajra@ubuntu-lab: ~/Lab5$ sudo apt update
[sudo] password for hajra:
Get:1 http://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,215 B]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,542 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,124 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 2,569 kB in 34s (76.0 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

`sudo apt upgrade -y`

```
hajra@ubuntu-lab: ~/Lab5$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  libzvb1-common libbson1 libavdevice60 libpostproc57 libavcodec60
  libzvb10t64 libavutil58 libswscale7 libswresample4 7zip libavformat60
  libavfilter9
Learn more about Ubuntu Pro at https://ubuntu.com/pro
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

2. Installed required packages for Docker and GUI support:

`sudo apt install ca-certificates curl gnupg lsb-release -y`

```
hajra@ubuntu-lab: ~/Lab5$ sudo apt install ca-certificates curl gnupg lsb-release -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.6).
gnupg is already the newest version (2.4.4-2ubuntu17.3).
gnupg set to manually installed.
lsb-release is already the newest version (12.0-2).
lsb-release set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```


3. Added Docker's official GPG key and repository:

```
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/etc/apt/keyrings/docker.gpg
hajra@ubuntu-lab: $ sudo mkdir -p /etc/apt/keyrings
hajra@ubuntu-lab: $ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
hajra@ubuntu-lab: $
```

Then add the Docker repository:

```
echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list >
/dev/null
```

```
hajra@ubuntu-lab: $ echo \
> "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
> https://download.docker.com/linux/ubuntu \
> $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

4. Updated apt and installed Docker packages (engine, CLI, and Desktop):

sudo apt update

```
hajra@ubuntu-lab: $ sudo apt update
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:3 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [33.3 kB]
Hit:4 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:7 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,541 kB]
Fetched 1,749 kB in 12s (143 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y

```
hajra@ubuntu-lab: $ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras libslirp0 pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libslirp0 pigz slirp4netns
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 105 MB of archives.
After this operation, 437 MB of additional disk space will be used.
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 1.7.28-1-ubuntu.24.04-noble [31.9 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libslirp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:28.5.1-1-ubuntu.24.04-noble [16.5 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:28.5.1-1-ubuntu.24.04-noble [19.7 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.29.1-1-ubuntu.24.04-noble [15.9 MB]
Get:8 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:28.5.1-1-ubuntu.24.04-noble [6,481 kB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 2.40.2-1-ubuntu.24.04-noble [14.3 MB]
Fetched 105 MB in 47s (2,228 kB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 187553 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_1.7.28-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking containerd.io (1.7.28-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5:28.5.1-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-cli (5:28.5.1-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5:28.5.1-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce (5:28.5.1-1-ubuntu.24.04-noble) ...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.29.1-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-buildx-plugin (0.29.1-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5:28.5.1-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:28.5.1-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../6-docker-compose-plugin_2.40.2-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-compose-plugin (2.40.2-1-ubuntu.24.04-noble) ...
```

5. Downloaded and installed Docker Desktop for Linux:

wget https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb

```
hajra@ubuntu-lab: $ wget https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb
--2025-10-25 16:35:58-- https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb
Resolving desktop.docker.com (desktop.docker.com)... 3.160.77.4, 3.160.77.45, 3.160.77.55, ...
Connecting to desktop.docker.com (desktop.docker.com)|3.160.77.4|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 421498404 (402M) [application/octet-stream]
Saving to: 'docker-desktop-amd64.deb'

docker-desktop-amd64.deb      100%[=====] 401.96M  1.97MB/s   in 3m 20s

2025-10-25 16:39:19 (2.01 MB/s) - 'docker-desktop-amd64.deb' saved [421498404/421498404]
```

sudo apt install ./docker-desktop-amd64.deb -y

```
hajra@ubuntu-lab: $ sudo apt install ./docker-desktop-amd64.deb -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'docker-desktop' instead of './docker-desktop-amd64.deb'
The following additional packages will be installed:
  cpu-checker ipxe-qemu ipxe-qemu-256k-compat-efi-roms libboost-iostreams1.83.0 libboost-thread1.83.0 libbrlapi0.8 libcacard0 libdaxctl1 libfdt1 libiscsi7 libndctl6 libpmem1 libpmemobj1
  librados2 librd1 librdmacm164 libspice-server1 libsubid4 liburing2 libusbredirparser164 libvirglrenderer1 msr-tools ovmf pass qemu-block-extra qemu-system-common qemu-system-data
  qemu-system-gui qemu-system-modules-opengl qemu-system-modules-spice qemu-system-x86 qemu-utils qrencode seabios tree uidmap wl-clipboard xcipx
Suggested packages:
  gststreamer1.0 libav gststreamer1.0-plugins-ugly libxml-simple-perl python ruby samba vde2
The following NEW packages will be installed:
  cpu-checker docker-desktop ipxe-qemu ipxe-qemu-256k-compat-efi-roms libboost-iostreams1.83.0 libboost-thread1.83.0 libbrlapi0.8 libcacard0 libdaxctl1 libfdt1 libiscsi7 libndctl6
  libpmem1 libpmemobj1 librados2 librd1 librdmacm164 libspice-server1 libsubid4 liburing2 libusbredirparser164 libvirglrenderer1 msr-tools ovmf pass qemu-block-extra qemu-system-common
  qemu-system-data qemu-system-gui qemu-system-modules-opengl qemu-system-modules-spice qemu-system-x86 qemu-utils qrencode seabios tree uidmap wl-clipboard xcipx
0 upgraded, 39 newly installed, 0 to remove and 0 not upgraded.
```

6. Started Docker Desktop application:

```
systemctl --user start docker-desktop
```

```
hajra@ubuntu-lab:~$ systemctl --user start docker-desktop
hajra@ubuntu-lab:~$
```

7. Verified Docker Desktop installation:

```
hajra@ubuntu-lab:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-10-25 16:34:38 UTC; 10min ago
   TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Main PID: 768054 (dockerd)
   Tasks: 0
   Memory: 21.7M (peak: 97.0M swap: 18.7M swap peak: 18.9M)
   CPU: 1.027s
   CGroup: /system.slice/docker.service
           └─768054 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 25 16:34:36 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:36.979378298Z" level=info msg="detected 127.0.0.53 nameserver, assuming system-resolved, so using resolv.conf: /run/systemd
Oct 25 16:34:37 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:37.056151433Z" level=info msg="Creating a containerd client" address=/run/containerd/containerd.sock timeout=1m0s
Oct 25 16:34:37 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:37.139148920Z" level=info msg="Loading containers: start."
Oct 25 16:34:38 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:38.110602097Z" level=info msg="Loading containers: done."
Oct 25 16:34:38 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:38.179496347Z" level=info msg="Docker daemon" commit=f8215cc containerd-snapshotter=false storage-driver=overlay2 version=20.5.1
Oct 25 16:34:38 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:38.180289680Z" level=info msg="Initializing buildkit"
Oct 25 16:34:38 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:38.257145265Z" level=info msg="Completed buildkit initialization"
Oct 25 16:34:38 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:38.270917845Z" level=info msg="Daemon has completed initialization"
Oct 25 16:34:38 ubuntu-lab dockerd[768054]: time="2025-10-25T16:34:38.271223404Z" level=info msg="API listen on /run/docker.sock"
Oct 25 16:34:38 ubuntu-lab systemd[1]: Started docker.service - Docker Application Container Engine.
```

- Opened Applications → Docker Desktop
- Confirmed the Docker icon appeared in the system tray.
- Verified that Docker Engine was running with:

```
docker version
```

```
hajra@ubuntu-lab:~$ docker version
Client: Docker Engine - Community
 Version: 20.5.1
 API version: 1.51
 Go version: go1.24.8
 Git commit: e180ab8
 Built: Wed Oct 8 12:17:26 2025
 OS/Arch: linux/amd64
 Context: default

Server: Docker Engine - Community
 Engine:
  Version: 20.5.1
  API version: 1.51 (minimum version 1.24)
  Go version: go1.24.8
  Git commit: f8215cc
  Built: Wed Oct 8 12:17:26 2025
  OS/Arch: linux/amd64
  Experimental: false
 containerd:
  Version: v1.7.28
  GitCommit: b9833aace56520842a23fa392a33f46af97866
 runc:
  Version: 1.3.0
  GitCommit: v1.3.0-0-g4ca628d1
 docker-init:
  Version: 0.19.0
  GitCommit: de40ad0
```

```
docker info
```

```
hajra@ubuntu-lab:~$ docker info
Client: Docker Engine - Community
 Version: 20.5.1
 Context: default
 Debug Mode: false
 Plugins:
  al: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version: v1.9.11
    Path: /home/hajra/.docker/cli-plugins/docker-ai
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.29.1-desktop.1
    Path: /home/hajra/.docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version: v2.40.2-desktop.1
    Path: /home/hajra/.docker/cli-plugins/docker-compose
  debug: Get a shell into any image or container (Docker Inc.)
    Version: 0.9.0
    Path: /home/hajra/.docker/cli-plugins/docker-debug
  desktop: Docker Desktop commands (Docker Inc.)
    Version: v0.2.0
    Path: /home/hajra/.docker/cli-plugins/docker-desktop
  extension: Manages Docker extensions (Docker Inc.)
    Version: v0.2.11
    Path: /home/hajra/.docker/cli-plugins/docker-extension
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version: v1.4.0
    Path: /home/hajra/.docker/cli-plugins/docker-init
  mcp: Docker MCP Plugin (Docker Inc.)
    Version: v0.24.0
    Path: /home/hajra/.docker/cli-plugins/docker-mcp
  offload: Docker Offload (Docker Inc.)
    Version: v0.5.1
    Path: /home/hajra/.docker/cli-plugins/docker-offload
  sandbox: Docker Sandbox (Docker Inc.)
    Version: v0.3.1
    Path: /home/hajra/.docker/cli-plugins/docker-sandbox
  shom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
    Version: 0.6.0
    Path: /home/hajra/.docker/cli-plugins/docker-shom
  scout: Docker Scout (Docker Inc.)
    Version: v1.10.3
    Path: /home/hajra/.docker/cli-plugins/docker-scout

Server:
 Containers: 0
  Running: 0
  Paused: 0
  Stopped: 0
 Images: 0
 Server Version: 20.5.1
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: systemd
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs jelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 runc
 Default Runtime: runc
 Init Binary: docker-init
 containerd version: b9833aace56520842a23fa392a33f46af97866
 runc version: v1.3.0-0-g4ca628d1
 init version: de40ad0
 Security Options:
  apparmor
  seccomp
   Profile: builtin
 cgroups:
  kernel: 0.8.0-86-generic
 operating system: ubuntu 24.04.3 LTS
 ostype: linux
 architecture: x86_64
 CPUs: 2
 Total Memory: 1.075GiB
 Name: ubuntu-lab
 ID: 49239b75-fc34-402c-82ab-b6e673409f02
 Docker Root Dir: /var/lib/docker
```

8. Install GUI (XFCE) Environment

Run this in your Ubuntu VM terminal:

```
sudo apt update
```

```
hajra@ubuntu-lab: ~$ sudo apt update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:7 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

```
sudo apt install xfce4 xfce4-goodies -y
```

```
hajra@ubuntu-lab: ~$ sudo apt install xfce4 xfce4-goodies -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
xfce4 is already the newest version (4.18).
xfce4-goodies is already the newest version (4.18.2build1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

This installs the XFCE desktop and some useful components (lightweight, works well in VMware).

It may take a few minutes.

9. Verify or (Re)Enable XRDP for Remote GUI Access

Now let's ensure you can actually open and *see* the XFCE desktop GUI through VMware (or RDP if needed).

Run this command next:

```
sudo systemctl enable --now xrdp
```

```
hajra@ubuntu-lab: ~$ sudo systemctl enable --now xrdp
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp
```

Then check XRDP status:

```
sudo systemctl status xrdp
```

```
hajra@ubuntu-lab: ~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-10-25 13:35:12 UTC; 3h 28min ago
     Docs: man:xrdp(8)
           man:xrdp.ini(5)
   Main PID: 1769 (xrdp)
    Tasks: 1 (limit: 2207)
   Memory: 148.0K (peak: 1.9M swap: 724.0K swap peak: 724.0K)
      CPU: 130ms
   CGroup: /system.slice/xrdp.service
           └─1769 /usr/sbin/xrdp

Oct 25 13:35:08 ubuntu-lab systemd[1]: Starting xrdp.service - xrdp daemon...
Oct 25 13:35:11 ubuntu-lab xrdp[1732]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 25 13:35:11 ubuntu-lab xrdp[1732]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 25 13:35:11 ubuntu-lab xrdp[1732]: [INFO ] xrdp_listen_pp done
Oct 25 13:35:11 ubuntu-lab systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after start: No such file or directory
Oct 25 13:35:12 ubuntu-lab systemd[1]: Started xrdp.service - xrdp daemon.
Oct 25 13:35:13 ubuntu-lab xrdp[1769]: [INFO ] starting xrdp with pid 1769
Oct 25 13:35:13 ubuntu-lab xrdp[1769]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 25 13:35:13 ubuntu-lab xrdp[1769]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 25 13:35:13 ubuntu-lab xrdp[1769]: [INFO ] xrdp_listen_pp done
```

10. Switch to GUI mode

Run this command in your terminal:

```
sudo systemctl set-default graphical.target
```

```
sudo systemctl isolate graphical.target
```

```
hajra@ubuntu-lab: ~$ sudo systemctl set-default graphical.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
hajra@ubuntu-lab: ~$ sudo systemctl isolate graphical.target
hajra@ubuntu-lab: ~$
```

Then, wait a few seconds — your screen should change to a **login window** (the XFCE GUI login).

If it doesn't, you can reboot manually:

```
sudo reboot
```

```
hajra@ubuntu-lab: ~$ sudo reboot
Broadcast message from root@ubuntu-lab on pts/1 (Sat 2025-10-25 17:11:38 UTC):

The system will reboot now!

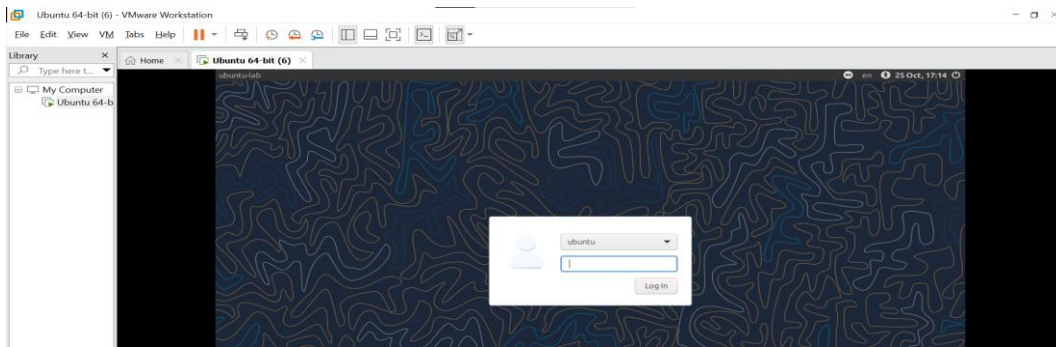
hajra@ubuntu-lab: ~$ client_loop: send disconnect: Connection reset
PS C:\Users\HVP>
```

11. Started the GUI session manually when in CLI mode:

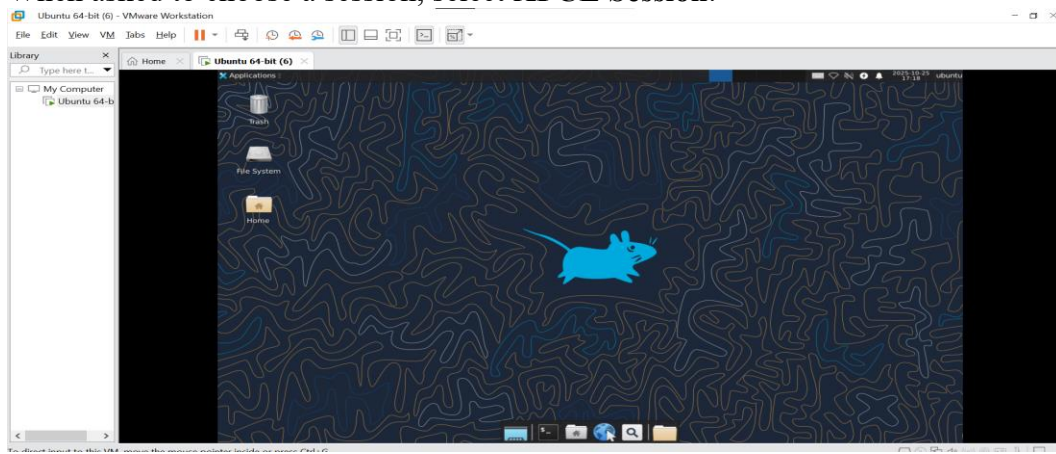
```
sudo systemctl start lightdm
```

After this:

- In VMware, your VM will show a **login screen** (username/password prompt).



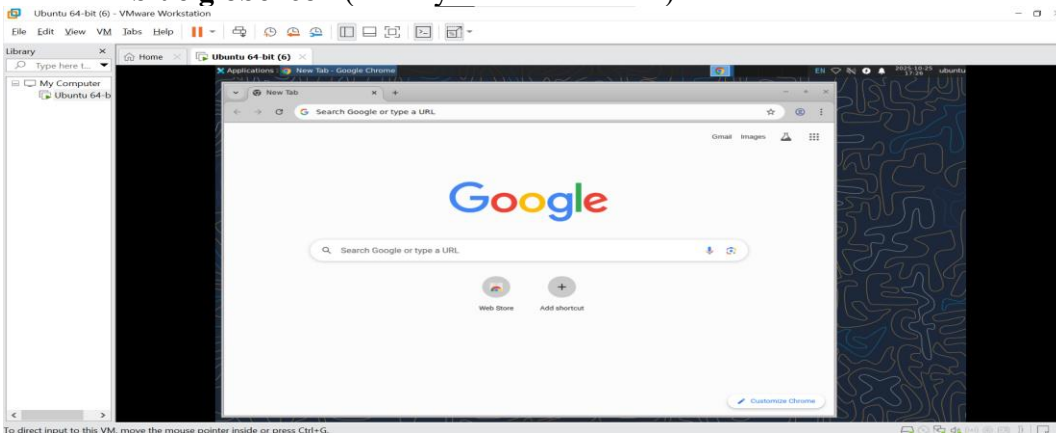
- Login as user **hajra**.
- When asked to choose a session, select **XFCE Session**.



12. Download the Docker Desktop .deb package

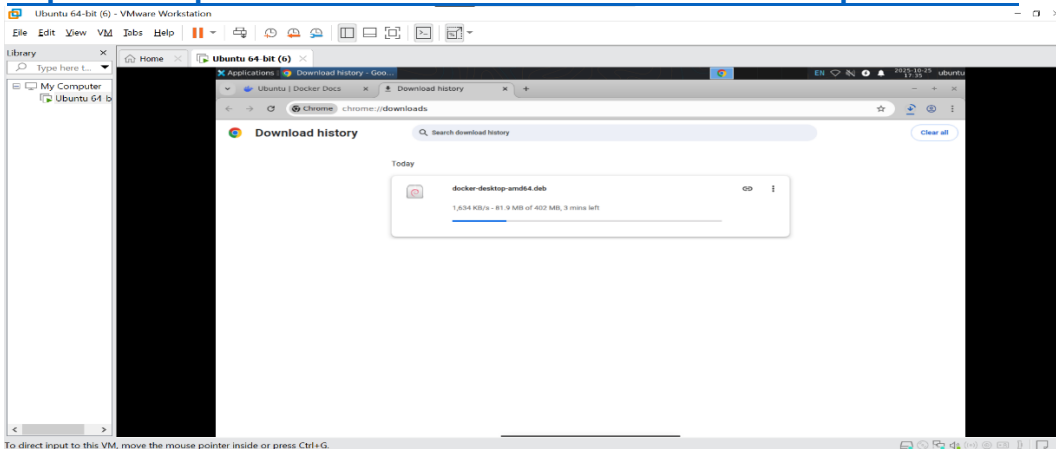
Now that you're inside XFCE:

1. Click the **blue globe icon** (that's your web browser).



2. In the browser, go to given link and this automatically download the file:

<https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb>



13. Install Docker Desktop

Run the following commands one by one **in your terminal** (you should still be in your home directory ~):

```
cd ~/Downloads
```

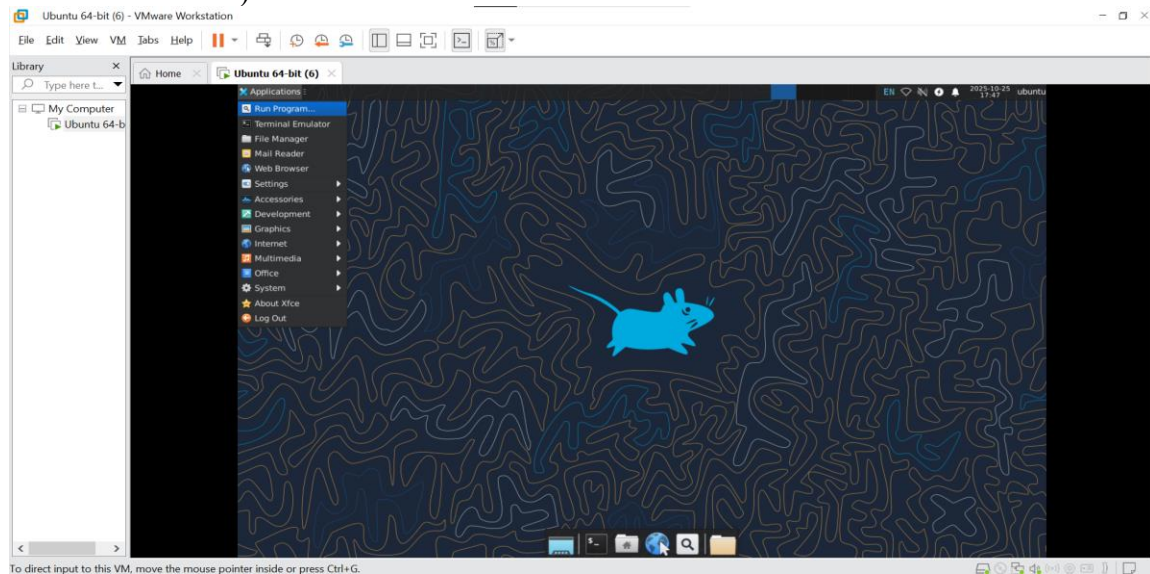
```
hajra@ubuntu-lab:~$ cd ~/Downloads
hajra@ubuntu-lab:~/Downloads$ ls
docker-desktop-amd64.deb
hajra@ubuntu-lab:~/Downloads$
```

Then install it using:

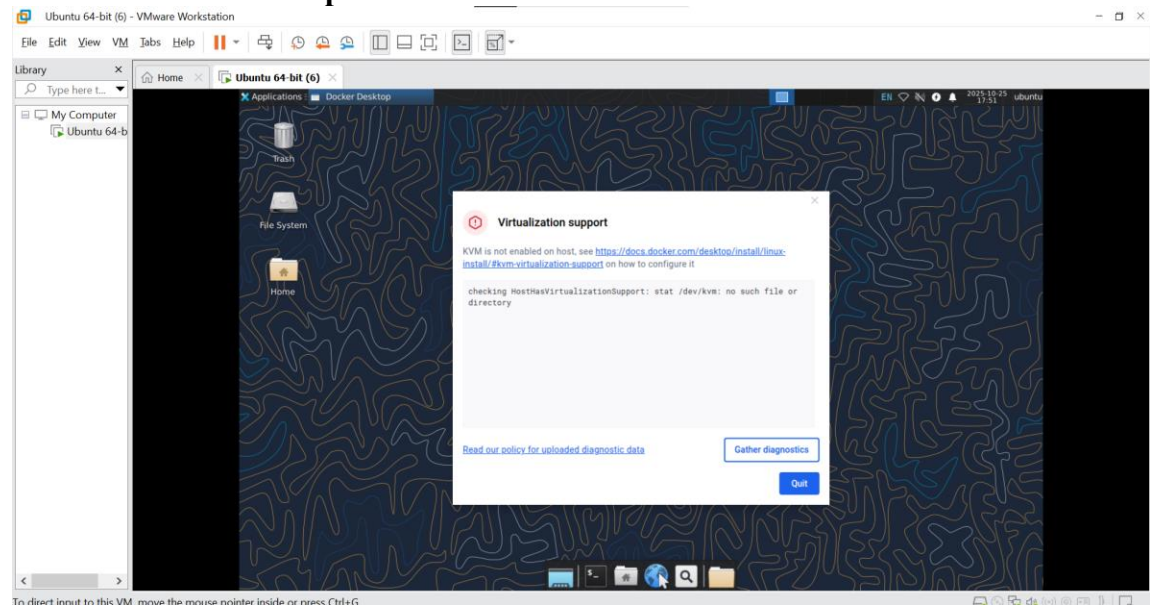
```
sudo apt install ./docker-desktop-amd64.deb -y
```

```
hajra@ubuntu-lab:~/Downloads$ sudo apt install ./docker-desktop-amd64.deb -y
[sudo] password for hajra:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'docker-desktop' instead of './docker-desktop-amd64.deb'
docker-desktop is already the newest version (4.49.0-208700).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
hajra@ubuntu-lab:~/Downloads$
```

1. In your **Ubuntu GUI** (XFCE desktop), click the **Applications** menu (usually at the top-left or bottom-left corner).



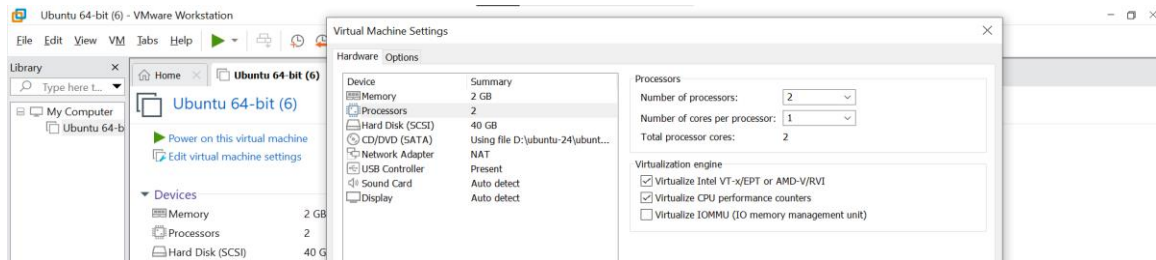
2. Type **Docker Desktop** in the search bar.
3. Click on **Docker Desktop** to launch it.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

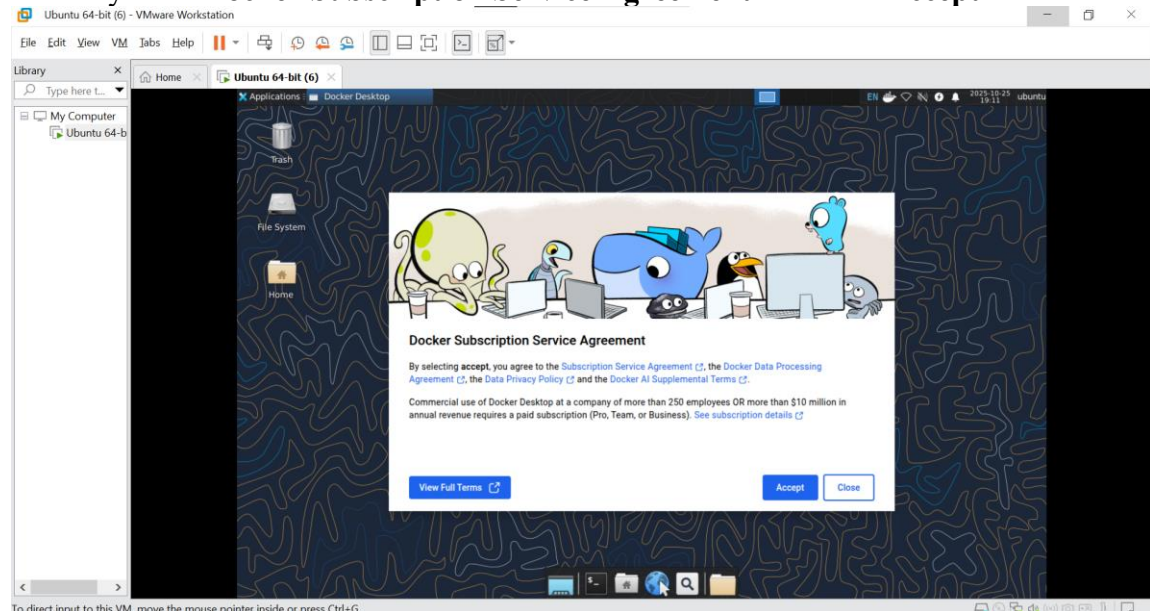
Virtualization Error.

Solution:



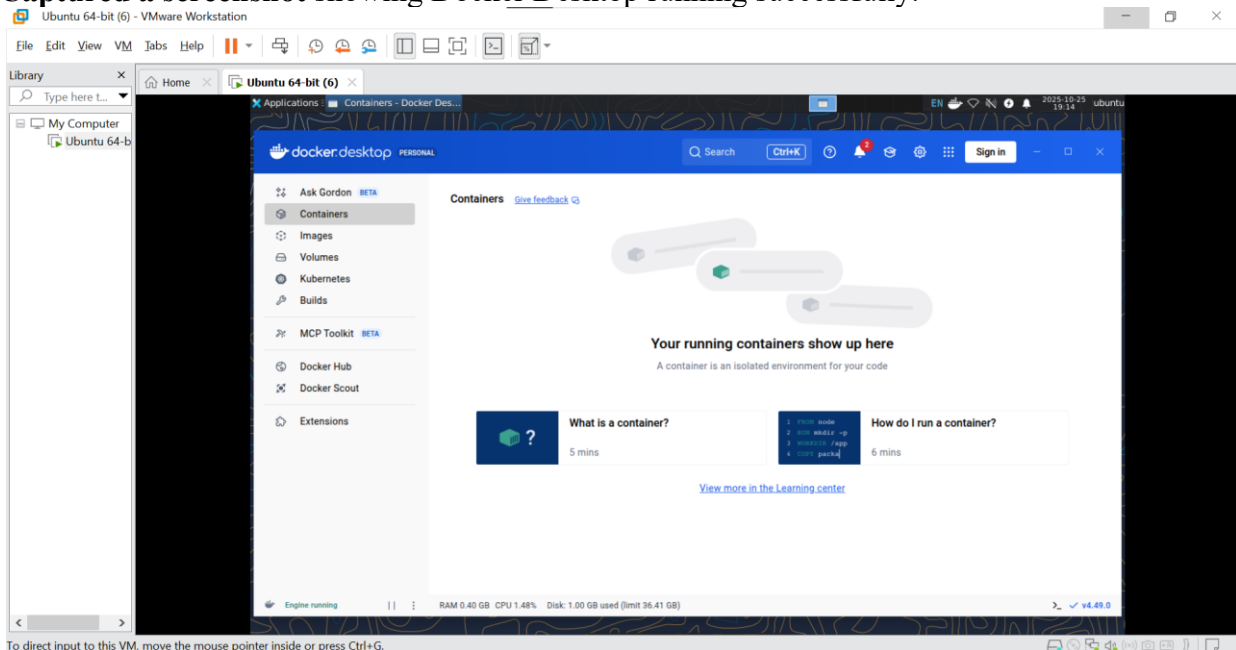
When it opens:

- You may see a **Docker Subscription Service Agreement** — click “Accept”.



- Then wait a bit while Docker Desktop starts up (it might take 30–60 seconds).
- Once you see the Docker whale 🐳 icon or the main Docker Desktop window, that means it's running.

14. Captured a screenshot showing Docker Desktop running successfully.



Result / Output

Docker Desktop was successfully installed and launched on the Ubuntu VM.

The verification confirmed that Docker Engine and Docker Desktop were both active and functional.